Statistical Analysis of Non-Profiling Higher-Order Distinguishers Against Inner Product Masking

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***Abstract*—Inner Product Masking (IPM) is one representative masking scheme, which captivates by so-called *Security Order Amplification* (SOA) property. It is commonly recognized that SOA holds under linear leakages. In this paper, we revisit SOA from a non-profiling attack perspective. Specifically, we conduct statistical analyses on three non-profiling distinguishers, including *Pearson Coefficient Distinguisher* (PCD), *Spearman Coefficient Distinguisher* (SCD) and *Kruskal-Wallis Distinguisher* (KWD). We find a fundamental connection between SCD and KWD such that SCD is a more generic distinguisher which encompasses KWD. Theoretical explanations for why KWD outperforms SCD under non-linear leakages are provided. We also propose a new adjusted SCD and present its optimal form, which bridges the efficiency gap with KWD. Grounded on this, SOA is extensively assessed and the observations are two-fold. On the one hand, we confirm again the effectiveness of SOA under Hamming weight leakage through the statistical analysis of PCD. On the other hand, we show that SOA can not resist rank-based distinguishers even under linear leakages, which has never been revealed before (to the best of our knowledge). At last, we verify the theoretical findings through both simulated and real-world measurements. Our results demonstrate the advantage of rank-based distinguishers in uncovering non-linear relationships hidden in leakage, enriching the tool-set for non-profiling class of side-channel attacks. Remarkably, we provide an adversary perspective to investigate SOA, highlighting that the side-channel resistance promised by SOA is vulnerable even considering the ideal linear leakage models.**

***Index Terms*—Inner product masking, security order amplification, non-profiling distinguishers, statistical analysis.**

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I. Introduction

SIDE-CHANNEL Attack (SCA) utilizes the observable physical leakage (such as executive time [1], power consumption [2], electromagnetic radiation [3] and so on) to disclose the sensitive information, thereby commonly regarded as one of the most threatening types of attacks against cryptographic implementations and devices. To thwart SCA, *masking* strategy, which offers provable security, has been a hot topic of research for decades in this field. The core idea of masking is to use a randomized function f_{enc} (termed as *encoding function*), that maps the sensitive variable Z to a set of random shares S_i with $1 \leq i \leq l$, such that $f_{\mathsf{enc}} : Z \rightarrow \{S_1, \ldots, S_l\}$. This randomized map reduces the association between algorithmic variable Z (that the adversary can simulate by public cryptographic algorithm) and actual intermediate values S_i during execution, thus enhancing side-channel security. Consequently, the security of masking is upper bounded by the encoding function f_{enc}, while *private computation* that proceeds encryption operations on the shares S_i offers the practical side-channel resistance.

A. Inner Product Masking and Security Order Amplification

Inner Product Masking (IPM) leverages inner product operation as encoding function, and captivates by *Security Order Amplification* (SOA) property [4], [5], quantifiable security level [6] and its nature to resist transitional leakage [7]. Regarding SOA of IPM, it was proposed in [4]: by simulating the mutual information of Inner Product (IP) encoding under Hamming Weight (HW) leakage model, Wang et al. [4] found that different IPM parameters present distinguishable security levels, even up to a gap of one security order in the probing model. Subsequently, SOA was further explained by connecting to bit-probing model in [5] and [6]. At this stage, SOA has been reinterpreted as the *bit-probing security* [5], [6] order of IP encoding can be higher than its *word-probing security* [8] under linear leakage. Informally speaking, SOA happens since IP encoding can mix more bit-wise shares than Boolean encoding if bit-wise shares leak independently. Hence it is recognized that SOA holds under linear leakages [5], [7]

Based on our knowledge, SOA can be defined from three different perspectives: 1) in the context of information theory, security order of an encoding function is connected to the slope of mutual information curves [4], [7] and SOA indicates that one's slope is lower than $-n$ with only n shares (see [4, Fig. 3] for example); 2) regarding theoretical security models, SOA is characterized by the bit-probing [5] order of IP encoding can be higher than its word-probing security order; 3) since security order indicates the order of attacks that the encoding (and ideally the private computation) can withstand, SOA can be explained from the non-profiling attack perspective that $(d+1)$-share encoding (usually with d-th probing order security) is resistant to $(d+t)$th-order attacks under linear leakage for $t \geq 1$ (see [9, Fig. 4] for example). Note that the first two are more on the defender's side, while the latter provides insight from the adversary's viewpoint.

B. Non-Profiling Attacks Against Inner Product Masking

Compared to profiling attack, which requires a copy of the target device for leakage distribution profiling, non-profiling attack comes to be more realistic. *Correlation Power Attack*[1] (CPA) is one representative non-profiling attack, that is extensively utilized in the community. Theoretically, [10] firstly proposes to quantify the attack efficiency of 2nd-order CPAs against Boolean Masking (BM) using statistical analysis, while [11] and [12] further conduct higher-order CPAs on higher-order moments against IPM (including BM). Empirically, [9], [11], [12] verify the effectiveness of SOA under the attack of higher-order CPA using real-world measurements. Although CPA is proved to be optimal if the side-channel leakage linearly depends on the hypothetical leakages [13], it is unable to capture subtle non-linear connections.

Some studies have explored non-profiling distinguishers that utilize non-linear components, while a limited body of work investigates their effectiveness in attacking IPM. For instance, *Spearman Coefficient Distinguisher* (SCD) is shown to outperform CPA by real SCA experiments [14] against unprotected implementations, but is never studied for IPM. Besides, Cristiani et al. [15] devise *Joint Moments Regression* (JMR) to defeat various masking schemes, especially including Boolean, affine and multiplicative masking schemes, while IPM stays untouched. Recently, a new non-profiling distinguisher leveraging *Kruskal-Wallis test* is proposed in [16] and verified by attack experiments to be more powerful than some non-parametric distinguishers (including *mutual information analysis* [17] and *Kolmogorov-Smirnov test-based distinguisher* [18]). However, this new distinguisher is used against Boolean masked implementations as well and lacks theoretical analysis for its effectiveness.

In conclusion, previous literature has demonstrated improvements in attack effectiveness mainly against Boolean masking by proposing non-profiling approaches that can leverage non-linear dependencies. In addition, some of them still lack theoretical analyses. As a consequence for IPM, it remains to investigate how SOA will perform under non-profiling distinguishers capable of extracting non-linear components.

C. Contribution

In this paper, we revisit SOA of IPM from a non-profiling attack perspective. In addition to *Higher-Order Pearson Coefficient Distinguisher* (HO-PCD) as in [11] and [12], we study another two non-profiling distinguishers which are based on ranked data, namely *Higher-Order Spearman Coefficient Distinguisher* (HO-SCD) and *Higher-Order Kruskal-Wallis Distinguisher* [16] (HO-KWD) against IPM.

[1]CPA in this paper specifically refers to the nonprofiling attacks using Pearson correlation coefficient as a distinguisher.

First, our study undertakes statistical analyses on both HO-SCD and HO-KWD, clearly delineating their fundamental theoretical linkage. Specifically, we find that HO-SCD is a more generic distinguisher that encompasses HO-KWD, in terms of attack effectiveness. As a bonus, we offer theoretical explanations for the comparative attack results between HO-SCD and HO-KWD as documented in [16]. To maximize attack efficiency, we advance the design of HO-SCD and propose a new non-profiling distinguisher denoted as *Adjusted Higher-Order Spearman Coefficient Distinguisher* (A-HO-SCD). Remarkably, we present the optimal form of A-HO-SCD, that bridges the efficiency gap with HO-KWD. Our theoretical findings are finally validated by both simulated and empirical experiments.

Second, we revisit SOA of IPM by lens of non-profiling distinguishers. On the one hand, we derive the condition for ineffective 2nd-order PCD against two-share IPM under HW leakage and thereby re-explain SOA theoretically from the perspective of statistical analysis. On the other hand, our analyses suggest that rank-based distinguishers (including HO-SCD, A-HO-SCD and HO-KWD) can effectively undermine SOA even under linear leakage scenarios, that is reported for the first time to the best of our knowledge. Regarding this phenomenon, theoretical explanations are provided through statistical analyses. These findings have also been corroborated through both simulated and empirical experiments.

In summary, our study makes a significant step toward addressing the previously unexplored area of attacking IPM using non-profiling distinguishers that effectively capture non-linear dependencies. We find that SOA can be promisingly compromised even under linear leakage when confronted with rank-based distinguishers, which is contrary to the common knowledge [5], [7]. In addition, our findings enrich the tool-set for the non-profiling class of SCA in evaluating the practical security of cryptographic implementations.

II. Preliminaries

In this section, we begin by presenting the notations. Next, we formalize non-profiling higher-order distinguishers used in this paper and define their respective efficiencies. For the sake of completeness, a brief description of IPM is also provided.

A. Notations

We denote the finite field of order q as $\mathbb{F}_q$. The upper letter X is used to denote a random variable (unless otherwise specified) with the lowercase letter x a particular element. The probability of the event $(X = x)$ is denoted by $P[X = x]$. The expectation (resp., standard deviation) of X is denoted as $\mathbb{E}(X)$ (resp., $\sigma(X)$). $\mathsf{Cov}(X, Y)$ computes the covariance between X and Y. Notably, we use $w_H(x)$ to denote HW of x over $\mathbb{F}_{2^n}$ and thus $w_H(x) = \sum_{i=1}^{n} x[i]$, where $x[i]$ represents the i-th bit of x. Let the symbols "$\oplus$" and "$\wedge$" denote the XOR and AND operation, respectively. Besides, the symbol "$\cdot$" represents field multiplication over $\mathbb{F}_q$ or the inner product of two vectors over $\mathbb{F}_q^l$ according to the context. Vectors are denoted as $\vec{X}$ over $\mathbb{F}_q^l$ with X_i the i-th element and $|\vec{X}|$ is the size of vector. Unless otherwise specified, let calligraphic letters (e.g., $\mathcal{C}$) denote a linear code. In this paper, two kinds of security orders are involved, namely probing (word-level) security order t_w and bit-probing (bit-level) security order t_b [6], that are defined in $\mathbb{F}_{2^n}$ and $\mathbb{F}_2$, respectively. In addition, we recall the following coding-theoretic definition [6].

Definition 1 (Dual Distance [19]) Considering a linear code $\mathcal{C}$, its dual distance $d_{\mathcal{D}}^{\perp}$ is the minimum Hamming weight $w_H(u)$ of nonzero $u \in \mathbb{F}^n$, such that $\sum_{c\in\mathcal{C}}(-1)^{c\cdot u} \neq 0$.

We present below two useful properties used in this paper to make it be self-contained.

Property 1: For every $z, m \in \mathbb{F}_{2^n}$, the HW of $z \oplus m$ satisfies

$$w_H(z \oplus m) = w_H(z) + w_H(m) - 2w_H(z \wedge m). \tag{1}$$

Property 2 (Lemma 6 of [10]) Let C and Z be two random variables. Then, for every function f defined over the support of Z, we have

$$\mathbb{E}(f(Z)C) = \mathbb{E}(f(Z)\mathbb{E}(C|Z)). \tag{2}$$

Lemma 1: Property 2 implies that for any function g defined over the support of Z, we have:

$$\mathbb{E}(f(g(Z))C) = \mathbb{E}(f(g(Z))\mathbb{E}(C|g(Z))). \tag{3}$$

Lemma 2 (Remark 7 of [10]) Property 2 implies $\mathbb{E}(C) = \mathbb{E}(\mathbb{E}(C|Z)) = \mathbb{E}(\mathbb{E}(C|g(Z)))$ (by setting $f : z \mapsto 1$).

Note that Lemma 2 is known as the law of total expectation.

B. Side-Channel Distinguishers

A side-channel distinguisher D exploits statistical tools to differentiate among various key guesses with predicted intermediates Z_k (associated with key guesses) and side-channel leakage L. Typically, it outputs distinguishing scores d_k and performs a maximization over all key guesses $k \in \mathbb{F}_q$ to find the most possible candidate $\dot{k}$. The attack succeeds if $\dot{k}$ is exactly the true key k^*, formalized as $\dot{k} = k^*$ where:

$$\dot{k} = \arg\max_{k\in\mathbb{F}_q} \{d_k\} = \arg\max_{k\in\mathbb{F}_q} \{\mathsf{D}(Z_k, L)\}. \tag{4}$$

In the presence of masking schemes, the key-dependent sensitive variable Z is split into l shares by an encoding function $f_{\mathsf{enc}} : Z \to \{S_1, \ldots, S_l\}$. Hence the adversary has to collect the respective l side-channel leakage $L_1, \ldots, L_l$ such that $L_i = \mathcal{L}(S_i) + N_i$ for $1 \leq i \leq l$, where $\mathcal{L}$ is the leakage function related to specific device and N_i denotes the noise. And then, the distinguisher D should combine those leakages L_i to launch a higher-order attack. In the sequel, we formally present the definition of Higher-Order Distinguisher (HOD).

Definition 2 (Higher-Order Distinguisher) Given a set of leakages $L_1, \ldots, L_l$ related to l shares of the sensitive variable Z, a higher-order distinguisher succeeds if $\dot{k} = k^*$ such that

$$\dot{k} = \arg\max_{k\in\mathbb{F}_q} \{d_k\} = \arg\max_{k\in\mathbb{F}_q} \{\mathsf{D}(Z_k, \mathsf{C}(L_1, \ldots, L_l))\}, \tag{5}$$

where C is a combining function that combines the side-channel leakages $L_1, \ldots, L_l$. The order of HOD is thus defined as the numeric degree of C denoted as $O = \deg(\mathsf{C})$.

In practice, distinguishers are estimators of statistical quantities. Thus in the formulas below we emphasize this fact by placing a wavy line above the respective quantity, such as $\tilde{\mathbb{E}}(X)$ in lieu of $\mathbb{E}(X)$. Assuming that the adversary acquires m measurements, it is foreseeable that $\tilde{d}_{k,m}$ tends toward d_k as m increases (by the law of large numbers). We give a definition for the efficiency of HOD, which is motivated by [10, Definition 3].

Definition 3 (Efficiency of HOD) The efficiency of HOD given a success rate β is the smallest m such that

$$P[\tilde{d}_{k^*,m} > \max_{k\in\mathbb{F}_q,\, k\neq k^*} \tilde{d}_{k,m}] \geq \beta. \tag{6}$$

As some distinguishers involve ranked data, we initially introduce the rank transformation. It proceeds as follows: consider a data set C and let C rank from the smallest to the largest, hence the smallest data has rank 1, the second smallest one has rank 2, etc. Ties are resolved by assigning the average of ranks that the ties would have received.

Definition 4 (Rank Transformation) Assume a data set C has a discrete distribution $(c_i, p_i)_i$ with $1 \leq i \leq v$, where (c_i, p_i) indicates that $P[C = c_i] = p_i > 0$ and $\sum_{i=1}^{v} p_i = 1$. For each $1 \leq i < j \leq v$, we assume $c_i < c_j$. After rank transformation of C, the distribution of the ranked data $\widehat{C}$ is $(\widehat{c_i}, p_i)_i$ for $1 \leq i \leq v$ where $\widehat{c_i}$ satisfies:

$$\widehat{c_i} = \sum_{j=1}^{i-1} p_j m + \frac{1}{2} p_i m + \frac{1}{2}, \tag{7}$$

where m is the size of data set such that $m = |C| = |\widehat{C}|$.

In the following, we formalize three non-profiling higher-order distinguishers investigated in this paper, notably two of which based on ranked data.

Definition 5 (Higher-Order Pearson Coefficient Distinguisher): HO-PCD computes the distinguishing score d_k by calculating the Pearson correlation coefficient between the simulated hypothetical leakage $\mathcal{H}(Z_k)$ and the combined side-channel leakage C, formalized as:

$$d_k = |\tilde{\rho}_k| = |\tilde{\rho}(\mathcal{H}(Z_k), C)|, \tag{8}$$

where $\mathcal{H}$ is the hypothetical leakage function simulated by adversary and $C = \mathsf{C}(L_1, \ldots, L_l)$. Pearson correlation coefficient ρ between two random variables X and Y is evaluated by:

$$\rho(X, Y) = \frac{\mathsf{Cov}(X, Y)}{\sigma(X)\sigma(Y)}. \tag{9}$$

Note that HO-PCD is exactly Higher-Order CPA [10], [20], e.g., defined as [12, Definition 3]. It rests on the following fact: if the hypothetical leakage function $\mathcal{H}$ and the combining function C are well constructed, then $\mathcal{H}(Z_{k^*})$ should be highly correlated to C, which leads to $|\tilde{\rho}_{k^*}|$ corresponding to the correct key much greater than $|\tilde{\rho}_k|$ for any other $k \in \mathbb{F}_q$ that $k \neq k^*$. We follow the assumption of [10] that there is a very low correlation between correct and incorrect key guesses, which implies that ρ_k are close to zero for $k \neq k^*$. Hence, the efficiency of HO-PCD mainly relies on d_{k^*}. This fact has also been argued in [10], [21], [22], and [23]. It is shown that the amount of measurements for a successful attack is around $\alpha/\rho_{k^*}^2$, where α is a value depending on the required success rate β and the size of key space $|\mathbb{F}_q|$.

Definition 6 (Higher-Order Spearman Coefficient Distinguisher): HO-SCD computes the distinguishing score d_k by calculating the Spearman correlation coefficient between the simulated hypothetical leakage $\mathcal{H}(Z_k)$ and the combined side-channel leakage C, formalized as:

$$d_k = |\tilde{\widehat{\rho}}_k| = |\tilde{\widehat{\rho}}(\mathcal{H}(Z_k), C)|, \tag{10}$$

where $\mathcal{H}$ is the hypothetical leakage function simulated by adversary and $C = \mathsf{C}(L_1, \ldots, L_l)$. Note that $\widehat{\rho}$ differs from ρ in that $\widehat{\rho}(X, Y) = \rho(\widehat{X}, \widehat{Y})$ with X and Y as random variables.

HO-SCD is a non-parametric alternative to HO-PCD, which is a direct application of HO-PCD on the ranked data. Hence, it can be inferred that the efficiency of HO-SCD depends on key-related distinguishing score d_{k^*} as well.

Definition 7 (Higher-Order Kruskal-Wallis Distinguisher) HO-KWD computes the distinguishing score as:

$$d_k = (m-1)\frac{\sum_{i=1}^{t} n_i(\tilde{\mathbb{E}}(\widehat{C}|g(Z_k)=i) - \tilde{\mathbb{E}}(\widehat{C}))^2}{\sum_{i=1}^{m}(\widehat{C}_i - \tilde{\mathbb{E}}(\widehat{C}))^2}, \tag{11}$$

where $g(Z_k)$ represents the method how $\widehat{C}$ is grouped. Specifically, $\{\widehat{C}|g(Z_k) = i\}$ contains the ranks of leakage samples satisfying $g(Z_k) = i$. Note that n_i denotes the size of group $\{\widehat{C}|g(Z_k) = i\}$ and thus $m = \sum_{i=1}^{t} n_i$, where t is the total number of groups.

For example, if $g(Z_k) = Z_k$ in Definition 7, then $\widehat{C}$ is grouped by the identity value of Z_k. As analyzed in [16], the null hypothesis is that all groups have the same distribution and thus $\tilde{\mathbb{E}}(\widehat{C}|g(Z_k) = i)$ should approach $\tilde{\mathbb{E}}(\widehat{C})$ as m increases. If $k \neq k^*$, the ranked leakage samples are randomly assigned to various groups, which leads to nearly same distributions for all groups and thereby a small d_k (ideally close to zero). As a consequence, the efficiency of HO-KWD primarily depends on d_{k^*}, that corresponds to true key.

Remark 1: In this paper, we optimize the attack efficiency of distinguishers by maximizing its key-related distinguishing score d_{k^*}, which is motivated by [10] and [12]. This is based on an ideal assumption that d_k for $k \neq k^*$ approaches zero, as is discussed above. We should underline that, in practice, the value of d_{k^*} is an important indicator of attack efficiency for distinguishers, but not the metric that fully measures the attack capability of distinguishers according to Definition 3.

C. Inner Product Masking

IPM involves two vectors $\vec{A}, \vec{B} \in \mathbb{F}_q^l$ with the encoding function defined as $f_{\mathsf{enc}} : Z \to \{S_1, S_2, \ldots, S_l\}$, where $S_1 = Z \oplus \sum_{i=2}^{l} B_i \cdot A_i$ and $S_i = B_i$ for $2 \leq i \leq l$. Note that $\vec{A}$ is public and fixed, while $\vec{B}$ is private and random. For efficient implementation, the first element of $\vec{A}$ is $A_1 = 1$. The remaining elements of $\vec{A}$, denoted as A_i for $i > 1$, are referred to as *IPM Parameters* [6].

The encoding form of IPM is expressed as:

$$\vec{Z} = \left(Z + \sum_{i=2}^{l} A_i \cdot B_i, B_2, B_3, \ldots, B_l\right) = Z\mathbf{G} + \vec{B}'\mathbf{H}, \tag{12}$$

where $\mathbf{G}$ and $\mathbf{H}$ are the generator matrices of linear codes $\mathcal{C}$ and $\mathcal{D}$, respectively. Note that $\vec{B}' = \{B_2, \ldots, B_l\}$.

The coding form of IPM allows to quantify the concrete security level of IPM by the coding theory. As proved in [6], the minimum distance $d_{\mathcal{D}}^{\perp}$ of the dual code $\mathcal{D}^{\perp}$ indicates the bit-probing security order of IP encoding.

III. Statistical Analysis of Distinguishers

In this section, we conduct statistical analyses on both HO-SCD and HO-KWD, and thereby reveal their relationship in theory. On this basis, an adjustment to HO-SCD is developed aimed at optimizing its efficiency. Hence, a new side-channel distinguisher denoted as *Adjusted HO-SCD* is proposed.

A. Relation Between HO-SCD and HO-KWD

In [10], Corollary 8 provides the optimal hypothetical leakage function and the optimal efficiency of HO-PCD through statistical analysis, which is recalled below:

Proposition 1 (See *[10, Corollary 8]*): *Let C be combined leakages and $\mathcal{H}(Z)$ be hypothetical leakages, HO-PCD has optimal hypothetical leakage function $\mathcal{H}_{opt}$ that maximizes distinguishing score as*:

$$\mathcal{H}_{opt}(z) = \mathbb{E}(C|Z = z). \tag{13}$$

Accordingly, the maximized distinguishing score d_{opt} is:

$$d_{opt} = \frac{\sigma(\mathbb{E}(C|Z))}{\sigma(C)}. \tag{14}$$

Since HO-SCD equals HO-PCD performed on ranked data, we conduct statistical analysis on HO-SCD motivated by Proposition 1. To begin with, we recall an important property proved in [10] and present two lemmas.

Property 3 (See *[10, Proposition 5]*): *Let C and Z be two random variables. Then for every function f over Z, we have*

$$\rho(f(Z), C) = \rho(f(Z), \mathbb{E}(C|Z)) \times \frac{\sigma(\mathbb{E}(C|Z))}{\sigma(C)}. \tag{15}$$

Lemma 3: Assuming that g is an arbitrary function over Z and $\mathcal{H}(Z) = f \circ g(Z)$, according to Property 3, we have

$$\rho(\mathcal{H}(Z), C) = \rho(\mathcal{H}(Z), \mathbb{E}(C|g(Z))) \times \frac{\sigma(\mathbb{E}(C|g(Z)))}{\sigma(C)}. \tag{16}$$

Lemma 4: Property 3 and Lemma 3 jointly imply that:

$$\widehat{\rho}(\mathcal{H}(Z), C) = \rho(\widehat{\mathcal{H}(Z)}, \mathbb{E}(\widehat{C}|g(Z))) \times \frac{\sigma(\mathbb{E}(\widehat{C}|g(Z)))}{\sigma(\widehat{C})}. \tag{17}$$

As analyzed in Section II-B, the efficiency of HO-SCD relies on $d_{k^*} = |\widehat{\rho}_{k^*}|$, and thus we have Proposition 1 below as a direct consequence of Lemma 4.

Corollary 1 (Optimal Efficiency of HO-SCD) Let $\widehat{C}$ be the ranked version of combined leakages C and g is an arbitrary *function over sensitive variable Z, the optimal efficiency of HO-SCD can be evaluated as*:

$$\widehat{d}_{opt} = \frac{\sigma(\mathbb{E}(\widehat{C}|g(Z)))}{\sigma(\widehat{C})}. \tag{18}$$

Proof: Since $0 \leq |\rho(\cdot)| \leq 1$, Lemma 4 implies that $d_{k^*} = |\widehat{\rho}_{k^*}| \leq \widehat{d}_{opt}$, which completes the proof. □

Recall that HO-KWD is a non-profiling distinguisher based on ranked data as well. In the following, we propose Theorem 1 to illustrate the theoretical connection between HO-KWD and HO-SCD by means of statistical analysis.

Theorem 1: Assume that $\widehat{d}_{k,opt}$ denotes the maximum distinguishing score of HO-SCD for each $k \in \mathbb{F}_q$, which satisfies Equation 18 (substituting Z by Z_k). Then the distinguishing score d_k of HO-KWD is:

$$d_k = (m-1)\widehat{d}^2_{k,opt}, \tag{19}$$

where m is the amount of side-channel measurements.

Proof: According to Definition 7 and Lemma 2, we have

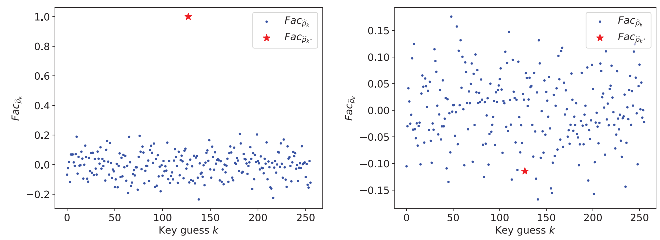
$$\begin{aligned} d_k &= (m-1)\frac{\sum_{y\in\mathcal{Y}} n_y(\mathbb{E}(\widehat{C}|g(Z_k)=y)-\mathbb{E}(\widehat{C}))^2}{\sum_{i=1}^{m}(\widehat{C}_i-\mathbb{E}(\widehat{C}))^2} \\ &= (m-1)\frac{\sum_{y\in\mathcal{Y}} \frac{n_y}{m}(\mathbb{E}(\widehat{C}|g(Z_k)=y)-\mathbb{E}(\mathbb{E}(\widehat{C}|g(Z_k))))^2}{\sum_{i=1}^{m}\frac{1}{m}(\widehat{C}_i-\mathbb{E}(\widehat{C}))^2} \\ &= (m-1)\frac{\sigma^2(\mathbb{E}(\widehat{C}|g(Z_k)))}{\sigma^2(\widehat{C})} \\ &= (m-1)\widehat{d}^2_{k,opt}, \end{aligned} \tag{20}$$

where $\mathcal{Y} \subseteq \mathbb{F}_q$ denotes the value space of $g(Z_k)$. □

Theorem 1 demonstrates that HO-SCD is equivalent to HO-KWD in terms of attack effectiveness if HO-SCD satisfies that $\mathsf{Fac}_{\widehat{\rho}_k} \triangleq \rho(\widehat{\mathcal{H}(Z_k)}, \mathbb{E}(\widehat{C}|g(Z_k)))$ are the same for each $k \in \mathbb{F}_q$. Hence, HO-SCD is a more generic distinguisher which covers HO-KWD as a special instance. Nevertheless, the advantage of HO-KWD is that it directly achieves the optimal attack efficiency $\widehat{d}_{opt}$ without the need to assume a proper hypothetical leakage function.

However, such feature also weakens the attack capability of HO-KWD. Note that HO-KWD allows each candidate key $k \in \mathbb{F}_q$ to directly reach the maximum distinguishing score $\widehat{d}^2_{k,opt}$. In other words, it essentially increases the distinguishing score d_k of each key candidate k to its maximum one $\widehat{d}_{k,opt}$ with underlying $\mathsf{Fac}_{\widehat{\rho}_k} = 1$. Thus the attack ability of HO-KWD depends on the gap between $\widehat{d}_{k^*,opt}$ and $\max_{k\in\mathbb{F}_q,\, k\neq k^*}\{\widehat{d}_{k,opt}\}$. While considering HO-SCD, a proper leakage hypothetical leakage function $\mathcal{H}(Z)$ will maximize $\mathsf{Fac}_{\widehat{\rho}_{k^*}}$ and minimize $\mathsf{Fac}_{\widehat{\rho}_k}$ (for $k \neq k^*$). Therefore, the practical capability of HO-SCD relies on the distinction between $(\mathsf{Fac}_{\widehat{\rho}_{k^*}} \times \widehat{d}_{k^*,opt})$ and $\max_{k\in\mathbb{F}_q,\, k\neq k^*}\{\mathsf{Fac}_{\widehat{\rho}_k} \times \widehat{d}_{k,opt}\}$.

Based on Theorem 1, the experimental results presented in [16] can be explained from a theoretical perspective. On the one hand, SCD is more powerful than KWD under HW leakage function (see Fig. 1(a) and Fig. 5(a) of [16]), since the $\mathcal{H}_{opt}(Z) = w_H(Z)$ utilized in [16] is indeed an appropriate hypothetical leakage function, that amplifies the difference



(a) HW leakage (b) SN leakage

Fig. 1. Simulated numerical evaluations of $\mathsf{Fac}_{\widehat{\rho}_k}$ for $k \in \mathbb{F}_{2^8}$. Note that sensitive variable Z is the first output byte of Sbox of AES without noise.

between $\mathsf{Fac}_{\widehat{\rho}_{k^*}}$ and $\max_{k\in\mathbb{F}_q,k\neq k^*}\{\mathsf{Fac}_{\widehat{\rho}_k}\}$, as displayed in Figure 1a. On the other hand, KWD outperforms SCD under randomly-weighted bits leakage and non-linear leakages, since SCD of [16] fails to simulate a proper hypothetical leakage function that enables $\mathsf{Fac}_{\widehat{\rho}_{k^*}} \geq \max_{k\in\mathbb{F}_q,k\neq k^*}\{\mathsf{Fac}_{\widehat{\rho}_k}\}$ in these scenarios. As a consequence, SCD cannot achieve as optimal attack efficiency $d_{k^*} = \widehat{d}_{opt}$ as KWD. Moreover, the gap between d_{k^*} and $\max_{k\in\mathbb{F}_q,\, k\neq k^*}\{d_k\}$ will be further reduced by larger $\max_{k\in\mathbb{F}_q,k\neq k^*}\{\mathsf{Fac}_{\widehat{\rho}_k}\}$, being smaller compared to that of KWD. As a simple proof, Figure 1b exhibits the numerical evaluation of $\mathsf{Fac}_{\widehat{\rho}_k}$ for $k \in \mathbb{F}_{2^8}$ under Strongly Non-linear (SN) leakage (refer to [16, Sec 4.1] or Section V-A of this paper to see the definition).

In summary, HO-SCD is fully equivalent to HO-KWD in terms of attack capability if its hypothetical leakage function $\mathcal{H}(Z_k)$ enables $\mathsf{Fac}_{\widehat{\rho}_k}$ to be identical among all the key candidates $k \in \mathbb{F}_q$. However, HO-SCD becomes more (resp., less) powerful if $\mathcal{H}(Z_k)$ makes $\mathsf{Fac}_{\widehat{\rho}_{k^*}} > \max_{k\in\mathbb{F}_q,k\neq k^*}\{\mathsf{Fac}_{\widehat{\rho}_k}\}$ (resp., $\mathsf{Fac}_{\widehat{\rho}_{k^*}} < \max_{k\in\mathbb{F}_q,k\neq k^*}\{\mathsf{Fac}_{\widehat{\rho}_k}\}$). Therefore, the hypothetical leakage function $\mathcal{H}(Z)$ plays a significant role in the attack effectiveness of HO-SCD. An optimal hypothetical leakage function should maximize $\mathsf{Fac}_{\widehat{\rho}_{k^*}}$ and meanwhile minimize $\max_{k\in\mathbb{F}_q,k\neq k^*}\{\mathsf{Fac}_{\widehat{\rho}_k}\}$ to bring out the best distinguishing ability of HO-SCD. In this paper, we optimize HO-SCD by maximizing $\mathsf{Fac}_{\widehat{\rho}_{k^*}}$ to obtain the optimal distinguishing score $\widehat{d}_{k^*} = \widehat{d}_{opt}$ (defined in Equation 18). Note that this aligns with the optimal efficiency of HO-SCD, that is discussed in Section II-B and defined in Corollary 1.

B. Adjusted HO-SCD

As analyzed above, to achieve optimal efficiency of HO-SCD, one should choose an appropriate hypothetical leakage function $\mathcal{H}(Z)$ enabling $\mathsf{Fac}_{\widehat{\rho}_{k^*}} = 1$. According to Lemma 4, $\widehat{\mathcal{H}(Z)} = \mathbb{E}(\widehat{C}|g(Z))$ should be obtained. However, due to the rank transformation, defining an appropriate function $\mathcal{H}(Z)$ that satisfies $\widehat{\mathcal{H}(Z)} = \mathbb{E}(\widehat{C}|g(Z))$ is much challenging. Interestingly, one simple solution exists, that is to adjust HO-SCD by taking $\mathcal{H}(Z) = \mathbb{E}(\widehat{C}|g(Z))$.

Definition 8 (Adjusted Higher-Order Spearman Coefficient Distinguisher (A-HO-SCD)): A-HO-SCD returns the distinguishing score $\widehat{d}_k$ as in:

$$\widehat{d}_k = |\tilde{\rho}(\mathcal{H}(Z_k), \widehat{C})|, \tag{21}$$

where $\widehat{C}$ is the ranked version of combined leakages C.

Corollary 2: Assume that $\widehat{C}$ is the ranked data of the combined leakages C. The optimal hypothetical leakage function $\mathcal{H}_{opt}$ of A-HO-SCD satisfies

$$\mathcal{H}_{opt}(z) = \mathbb{E}(\widehat{C}|g(z)). \tag{22}$$

Accordingly, the optimal efficiency $\widehat{d}_{opt}$ of A-HO-SCD is:

$$\widehat{d}_{opt} = \frac{\sigma(\mathbb{E}(\widehat{C}|g(Z)))}{\sigma(\widehat{C})}. \tag{23}$$

We should highlight that A-HO-SCD is a new side-channel distinguisher, which differs from HO-SCD since the hypothetical leakage $\mathcal{H}(Z_k)$ requires no rank transformation. This adjustment is made for enabling $\mathsf{Fac}_{\rho_{k^*}} \triangleq \rho(\mathcal{H}(Z_k), \mathbb{E}(\widehat{C}|g(Z_k))) = 1$ to achieve the optimal efficiency of A-HO-SCD (as well as HO-SCD). Hence from the perspective of statistical analysis, A-HO-SCD with the optimal hypothetical leakage function is an enhancement of HO-SCD regarding the attack efficiency considered in this paper (the key-related distinguishing score d_{k^*}). We also highlight that Theorem 1 can be applied directly to A-HO-SCD.

IV. Distinguishers Against Inner Product Masking

In this section, we investigate the attack ability of A-HO-SCD (also HO-SCD) and HO-KWD against IPM through statistical analysis. As a baseline, we include the *2nd-Order PCD* (2O-PCD) first and analyze its efficiency against IPM.

A. Leakage Model

Assume that $Z = f(P, k^*)$ is the sensitive variable over $\mathbb{F}_{2^n}$ (we instantiate $\mathbb{F}_q$ as $\mathbb{F}_{2^n}$ in the remainder of this paper), where P is the public variable that the adversary can access. Since Z is protected by l-share IPM, there is $f_{\mathsf{enc}} : Z \to \{S_1, \ldots, S_l\}$ over $\mathbb{F}_{2^n}$. Let the leakage function be $L(x) = \mathcal{L}(x) + N$, hence we have $L_i = \mathcal{L}(S_i) + N_i$ for each share S_i where N_i are i.i.d for $1 \le i \le l$. To launch higher-order attack, one should define a combining function C such that $C = \mathsf{C}(L_1, \ldots, L_l)$ with $\deg(\mathsf{C}) \ge l$.

B. Second-Order PCD Against IPM

In this subsection, we consider two-share IPM such that $S_1 = Z \oplus (A_2 \cdot B_2)$ and $S_2 = B_2$ satisfying $Z = S_1 \oplus (A_2 \cdot S_2)$. For brevity, we denote $a = A_2$ and $M = B_2$. The leakage function is instantiated as HW leakage model and thus $L(x) = w_H(x) + N$ with N following Gaussian distribution such that $N \sim \mathcal{N}(0, \sigma^2)$. Hence for two shares S_1 and S_2, the corresponding leakage are:

$$\begin{aligned} L_1 &= w_H(Z \oplus a \cdot M) + N_1, \\ L_2 &= w_H(M) + N_2. \end{aligned} \tag{24}$$

To combine both leakages L_1 and L_2, the centered product combining function is exploited such that $C = \mathsf{C}(L_1, L_2) = (L_1 - \mathbb{E}(L_1))(L_2 - \mathbb{E}(L_2))$. This combining function is proved to be optimal for two-share BM under HW leakage [10]

TABLE I

Optimal Efficiency of Various Two-Share IPM Instances

IPM instances	a	$d_{\mathcal{D}}^{\perp}$	t_w	t_b	d_{opt}
IPM1	1	2	1	1	0.3542
IPM2	2	2	1	1	0.2801
IPM3	3	3	1	2	0
IPM14	14	3	1	2	0
IPM23	23	4	1	3	0

and is widely used for higher-order attacks [11], [12], [16]. Consequently, the combined leakage C is:

$$\begin{aligned} C &= w_H(z \oplus a \cdot M) \times w_H(M) + w_H(z \oplus a \cdot M) \times N_2 \\ &\quad + w_H(M) \times N_1 + N_1 N_2 - \frac{n}{2} w_H(M) \\ &\quad - \frac{n}{2} w_H(z \oplus a \cdot M) - \frac{n}{2} N_1 - \frac{n}{2} N_2 + \frac{n^2}{4}. \end{aligned} \tag{25}$$

As claimed in Proposition 1, HO-PCD should simulate a perfect hypothetical leakage function $\mathcal{H}_{opt} = \mathbb{E}(C|Z)$ to attain the optimal efficiency. Below we derive the optimal hypothetical leakage function $\mathcal{H}_{opt}$ on the basis of Equation 25.

Theorem 2 (Optimal Hypothetical Leakage Function of 2O-PCD under HW Leakage): Let $Z \in \mathbb{F}_{2^n}$ be the sensitive variable which is protected by two-share IPM with a as IPM parameter. Then we have

$$\begin{aligned} \mathbb{E}(C|Z = z) = \frac{n}{2} w_H(z) - 2 \left(\sum_{i=1}^{n} z[i] \sum_{j=1}^{n} P_{i,j} \right) \\ + \mathbb{E}(w_H(a \cdot M) w_H(M)) - \frac{n^2}{4}, \end{aligned} \tag{26}$$

where $P_{i,j} = P[(a \cdot M)[i] = 1, M[j] = 1]$. For example, if $a = 1$, then we have $P_{i,i} = \frac{1}{2}$ and $P_{i,j} = \frac{1}{4}$ for $i \neq j$.

The detailed proof of Theorem 2 is shown in Appendix A. According to Theorem 2, we can obtain the following lemma.

Lemma 5: If $(a \cdot M)[i]$ is independent of $M[j]$ for $1 \le i, j \le n$, then we have $d_{opt} = 0$.

Proof: If $(a \cdot M)[i]$ is independent of $M[j]$ for $1 \le i, j \le n$, then we have $P_{i,j} = \frac{1}{4}$ for $1 \le i, j \le n$, hence we get

$$\mathbb{E}(C|Z = z) = \mathbb{E}(w_H(a \cdot M) w_H(M)) - \frac{n^2}{4}, \tag{27}$$

leading to $\sigma(\mathbb{E}(C|Z)) = 0$ since $\mathbb{E}(C|Z)$ does not depend on Z. Hence, one gets $d_{opt} = 0$. □

Lemma 5 demonstrates that IPM parameter possesses the capability to eliminate the secret-dependent information contained in $\mathbb{E}(C|Z)$, leading to $d_{opt} = 0$, which further indicates that 2O-PCD turns to be ineffective. Hence it provides another perspective to explain SOA of IPM: there exist some values of IPM parameters rendering $d_{opt} = 0$, thereby enabling two-share IPM to withstand 2O-PCD, which showcases SOA from an attack perspective. To exhibit the connection between our analysis and the coding-theory-based quantification framework of IPM presented in [6], Table I displays the optimal efficiency d_{opt} and the dual distance $d_{\mathcal{D}}^{\perp}$ of IP encoding for various two-share IPM instances (where IPMx denotes the instance with $a = x$ for $x \in \{1, 2, 3, 14, 23\}$). Note that the value of d_{opt} is derived directly from Equations 14, 25 and 26 using $Z \in \mathbb{F}_{2^8}$

without considering the noise N for simplicity. It is indicated in Table I that SOA emerges (when bit-probing security order t_b is higher than probing security order t_w) among the IPM instances with $d_{opt} = 0$. Note that such conclusion is not limited to the instances presented in Table I, but applicable for all $a \in \mathbb{F}_{2^8}\backslash\{0\}$. Therefore, SOA of two-share IPM under HW leakage is confirmed again from an attack perspective by means of statistical analysis.

Remark 2: A recent work [12] conducts statistical analysis on higher-order PCD against two-share IPM as well, but at higher orders than merely 2. Although motivated by the same idea in [10] and coming to the same conclusion, [12] applies coding theory to derive the condition for ineffective and optimal high-order CPAs [12, Theorems 5&6], while we follow the statistical analysis by probability theory. In particular, we focus on the 2O-PCD in this work as a comparison baseline.

C. Rank-Based Distinguishers Against IPM

Assume that the combined leakage samples set $C = \mathsf{C}(L_1, \ldots, L_l)$ has the distribution $(c_i, p_i)_i$[2] with $1 \le i \le v$, where (c_i, p_i) denotes that $P[C = c_i] = p_i > 0$ and $\sum_{i=1}^{v} p_i = 1$. For each $1 \le i < j \le v$, we assume $c_i < c_j$. Accordingly, the distribution of the ranked data $\widehat{C}$ is $(\widehat{c}_i, p_i)_i$ for $1 \le i \le v$, where $\widehat{c}$ is represented as in Equation 7. We define that for the leakage samples subset $\{\widehat{C}|g(Z) = y\}$ with $y \in \mathcal{Y} \subseteq \mathbb{F}_{2^n}$, its distribution satisfies $(\widehat{c}_i, p_i^y)_i$ for $1 \le i \le v$. Note that $p_i^y \ge 0$ for $1 \le i \le v$ and $\sum_{i=1}^{v} p_i^y = 1$. We denote $\vec{p} = (p_1, \ldots, p_v)$ and $\vec{p^y} = (p_1^y, \ldots, p_v^y)$.

As analyzed in Corollary 2, A-HO-SCD requires a perfect hypothetical leakage function for optimal attack efficiency. Hence we derive Theorem 3, which addresses the optimal hypothetical leakage function of A-HO-SCD under any leakage function against IPM at any order. Recall that HO-KWD obviates the simulation for such a hypothetical function.

Theorem 3 (Optimal Hypothetical Leakage Function of A-HO-SCD against IPM at Any Order): Take the above setting for the combined leakage $\widehat{C}$ and assume that $p_i^* = 2\sum_{j=1}^{i-1} p_j + p_i$ for $1 \le i \le v$ with the corresponding vector $\vec{p^*} = (p_1^*, \ldots, p_v^*)$. Let $y = g(z)$, then the optimal hypothetical leakage function of A-HO-SCD is:

$$\mathcal{H}_{opt}(z) = A \circ (\vec{p^*} \cdot \vec{p^y}), \tag{28}$$

where A is the affine function defined over $(\vec{p^*} \cdot \vec{p^y})$.

Proof: By Corollary 2, the optimal hypothetical leakage function is $\mathcal{H}_{opt}(z) = \mathbb{E}(\widehat{C}|g(z))$. Taking Equation 7, we have

$$\mathbb{E}(\widehat{C} \mid g(z) = y) = \sum_{i=1}^{v} p_i^y \widehat{c}_i = \frac{m}{2}\left(\sum_{i=1}^{v}\sum_{j=1}^{i-1} 2p_j p_i^y + \sum_{i=1}^{v} p_i p_i^y\right) + \frac{1}{2} = \frac{m}{2}\vec{p^*} \cdot \vec{p^y} + \frac{1}{2}. \tag{29}$$

□

Lemma 6: $\vec{p^*} \cdot \vec{p} = 1$.

Proof: According to Theorem 3, we have

$$\vec{p^*} \cdot \vec{p} = \sum_{i=1}^{m}\sum_{j=1}^{i-1} 2p_j p_i + \sum_{i=1}^{m} p_i^2 = \left(\sum_{i=1}^{m} p_i\right)^2 = 1. \tag{30}$$

□

We should underline that the optimal hypothetical leakage function of A-HO-SCD derived in Theorem 3 adapts to any leakage function $\mathcal{L}$ and any order of IPM. This adaptability stems from its utilization of the distribution of actual side-channel measurements, which closely aligns with the characteristics of the ranked leakage set. More interestingly and notably, the critical part $\mathsf{Fac}_{\rho_{k^*}}$ comprising the distinguishing score $\widehat{d}_{k^*}$ will not be affected by the noise levels. The distinction between HO-PCD and A-HO-SCD primarily arises from the difference between c_i and $\widehat{c}_i$. The former is a specific value associated with the sensitive variable, whereas the latter pertains to the leakage distribution.

Remark 3: The optimal hypothetical leakage function derived in Theorem 3 enables $\mathsf{Fac}_{\rho_{k^*}} = 1$, yet meanwhile allows that $\mathsf{Fac}_{\rho_k} = 1$ for each key guess $k \in \mathbb{F}_{2^n}$. As a consequence, A-HO-SCD with hypothetical leakage in this form becomes equivalently to HO-KWD according to Theorem 1.

As proposed in Theorem 1 and Corollary 2, the optimal efficiency of A-HO-SCD (also HO-SCD) is $\widehat{d}_{opt}$ (represented in Equation 23) and recall that the efficiency of HO-KWD relates to $\widehat{d}_{opt}$ as well. Next, we derive its specific form against IPM at any order under any leakage function.

Theorem 4 (Optimal Efficiency of A-HO-SCD and HO-SCD against IPM at Any Orders): Taking the above setting for combined leakage $\widehat{C}$, we have

$$\widehat{d}_{opt} = \sqrt{\frac{\mathbb{E}((\vec{p^*} \cdot \vec{p^y})^2) - 1}{\sum_{i=1}^{v} p_i (p_i^*)^2 - 1}}. \tag{31}$$

Please see Appendix B for the detailed proof.

Lemma 7: if there exists $y \in \mathcal{Y} \subseteq \mathbb{F}_{2^n}$ such that $\vec{p^*} \cdot \vec{p^y} \neq 1$, then we have $\widehat{d}_{opt} > 0$.

Proof: Let $e_i^y = p_i - p_i^y$ for $y \in \mathcal{Y}$ and $1 \le i \le v$. We denote that $\vec{e^y} = (e_1^y, \ldots, e_v^y)$ and thus we have:

$$\mathbb{E}(\widehat{C}|g(Z) = y) = \frac{m+1}{2} - \frac{m}{2}\vec{p^*} \cdot \vec{e^y}. \tag{32}$$

According to Equation 32, $\mathbb{E}(\mathbb{E}(\widehat{C}|g(Z)))$ satisfies

$$\begin{aligned}\mathbb{E}(\mathbb{E}(\widehat{C}|g(Z))) &= \frac{m+1}{2} - \frac{m}{2}\mathbb{E}(\vec{p^*} \cdot \vec{e^y}) \\ &= \frac{m+1}{2} - \frac{m}{2}\sum_{y \in \mathcal{Y}} P[g(Z) = y](\vec{p^*} \cdot \vec{e^y}).\end{aligned} \tag{33}$$

Since by Lemma 2: $\mathbb{E}(\mathbb{E}(\widehat{C}|g(Z))) = \mathbb{E}(\widehat{C}) = \frac{m+1}{2}$, we have

$$\frac{m}{2}\sum_{y \in \mathcal{Y}} P[g(Z) = y](\vec{p^*} \cdot \vec{e^y}) = 0. \tag{34}$$

Note that $P[g(Z) = y] > 0$. If $\widehat{d}_{opt} = 0$, there is $\sigma(\mathbb{E}(\widehat{C}|g(Z))) = 0$, then for each pair $(y_1, y_2) \in \mathcal{Y} \times \mathcal{Y}$, we have $\mathbb{E}(\widehat{C}|g(Z) = y_1) = \mathbb{E}(\widehat{C}|g(Z) = y_2)$ and thereby $\vec{p^*} \cdot \vec{e^{y_1}} = \vec{p^*} \cdot \vec{e^{y_2}}$. According to Equation 34, it implies that $\vec{p^*} \cdot \vec{e^y} = 0$ for each $y \in \mathcal{Y}$.

[2]Although noise N is a continuous variable over $\mathbb{R}$, the acquisition stage in the context of SCA will render the leakages C to be discrete.

TABLE II
NUMERICAL EVALUATIONS OF OPTIMAL EFFICIENCY AND DUAL DISTANCE OF VARIOUS TWO-SHARE IPM INSTANCES

IPM instances	a	$d_{\mathcal{D}}^{\perp}$	t_w	t_b	$\widehat{d}_{opt}$	$\widehat{d}'_{opt}$
IPM1	1	2	1	1	0.3200	0.3194
IPM2	2	2	1	1	0.2529	0.1996
IPM3	3	3	1	2	0.0288	0.0145
IPM14	14	3	1	2	0.0292	0.0111
IPM23	23	4	1	3	0.0252	0.0038

Hence for each $y \in \mathcal{Y}$, we have $\vec{p^*} \cdot \vec{p} = \vec{p^*} \cdot \vec{p^y}$. Since $\vec{p^*} \cdot \vec{p} = 1$ as proved in Lemma 6, we get $\vec{p^*} \cdot \vec{p^y} = 1$ for each $y \in \mathcal{Y}$, which contradicts the hypothesis as claimed in Lemma 7. This concludes the proof. □

Lemma 7 suggests that if there exists $y \in \mathcal{Y}$ enabling $\vec{p^*} \cdot \vec{p^y} \neq 1$, HO-SCD, A-HO-SCD and HO-KWD are all capable of capturing the sensitive information concealed within $\mathbb{E}(\widehat{C}|g(Z))$. Consequently, those rank-based distinguishers hold promise for effectively compromising IPM. We now turn to examine the conditions under which an attack would be ineffective (or saying $\widehat{d}_{opt} = 0$). That is, for each $y \in \mathcal{Y}$, there should be $\vec{p^*} \cdot \vec{p^y} = 1$ and $\vec{p^y} \neq \vec{p}$ (recall that $\vec{p^*} \cdot \vec{p} = 1$). Consider an extreme scenario where $\deg(\mathsf{C}) < l$. In this case, the combined leakage C is not associated with the sensitive variable $g(Z)$ guaranteed by the IP encoding. Consequently. the respective $\vec{p^y}$ should converge towards $\vec{p}$, resulting in $\vec{p^y} = \vec{p}$ and thereby $\vec{p^*} \cdot \vec{p^y} = 1$. This corresponds with the prevailing perceptions of higher-order attacks, which stipulate that an attack will fail if it has $\deg(\mathsf{C}) < l$. However considering the case where $\deg(\mathsf{C}) \geq l$, satisfying the above-mentioned condition for ineffective attack is challenging since $\vec{p^y}$ and $\vec{p}$ (determining $\vec{p^*}$) depend on several factors, including IPM parameters $\vec{A}$, leakage function $\mathcal{L}$, the distribution of noise N and the specific combining function C.

An interesting takeaway of Lemma 7 is that, even under linear leakage $\mathcal{L}$ (e.g., HW leakage), rank-based distinguishers demonstrate capability to extract information related to sensitive variable Z from noisy measurements $\widehat{C}$, despite the protection of SOA. For instance, 2nd-Order SCD (2O-SCD), Adjusted 2nd-Order SCD (A-2O-SCD) and 2nd-Order KWD (2O-KWD) are all promising to attack two-share IPM successfully due to $\widehat{d}_{opt} > 0$, even if those two shares S_1 and S_2 are leaked in a linear function. Note that this indicates that SOA of IPM is promising to be broken even under linear leakage from an attack perspective. To exhibit the contrast with HO-PCD shown in Table I, Table II provides the numerical evaluations[3] (using 100,000 simulated measurements) of optimal efficiency of A-2O-SCD (also 2O-SCD) and 2O-KWD against various two-share IPM instances. All conditions employed to derive $\widehat{d}_{opt}$ are consistent with those specified in Table I, including HW leakage function $\mathcal{L}(Z) = w_H(Z)$ without noise, the centered product combining function $\mathsf{C}(L_1, L_2) = (L_1 - \mathbb{E}(L_1)) \times (L_2 - \mathbb{E}(L_2))$ and IPM parameter $a \in \{1, 2, 3, 14, 23\}$. We choose $g(Z) = Z$ for the third column $\widehat{d}_{opt}$ and $g(Z) = w_H(Z)$ for the fourth column $\widehat{d}'_{opt}$. It is indicated in Table II that the optimal efficiency $\widehat{d}_{opt}$ and $\widehat{d}'_{opt}$ for all the considered two-share IPM instances are larger than zero, implying the existence of at least $(t_b + 1)$-th order leakages hidden in $\mathbb{E}(\widehat{C}|g(Z))$, which can be leveraged by rank-based distinguishers. It should be highlighted that this scenario actually holds for all $a \in \mathbb{F}_{2^8}\backslash\{0\}$ in IPM. Therefore, if statistical analyses in Section IV-B allow us to re-affirm the existence of SOA under conditions of linear leakage, then Lemma 7 and results of Table II indicate that from the attacker's viewpoint, SOA can not resist rank-based distinguishers even under HW leakage functions.

The advantage of rank-based distinguishers over HO-PCD lies in the difference between c_i and $\widehat{c}_i$. Precisely, c_i is a direct image of C with the input value $z \in \mathcal{Z}_i$, while $\widehat{c}_i$ relates to $\{p_1, p_2, \ldots, p_i\}$ involving subsets $\mathcal{Z}_1 \cup \mathcal{Z}_2 \cup \cdots \cup \mathcal{Z}_i \subseteq \mathcal{Z}$, where Z_i indicates a set $z \in \mathcal{Z}_i$ such that $c_i \leftarrow \mathsf{C} \circ L \circ \mathcal{L} \circ f_{\mathsf{enc}}(z)$. In other words, the rank transformation enables $\widehat{c}_i$ to be more informative than c_i and thus enhance the attack efficiency of rank-based distinguishers. However, we should underline that $\widehat{d}_{opt} > 0$ does not necessarily indicate a successful attack. Nevertheless, $\widehat{d}_{opt} > 0$ indicates that the information related to sensitive variable Z has been leaked and thereby can be leveraged by the distinguisher, a scenario markedly distinct from cases where $d_{opt} = 0$ in theory.

Remark 4: In fact, we do not impose constraints on the construction of $\widehat{C}$ (recall in Theorems 3 and 4), therefore the optimal attack efficiency and optimal hypothetical leakage function should extend to multiplicative masking as well. As of affine masking, however, it requires more careful investigation because of the mixture nature of both Boolean and multiplicative maskings. We leave the investigation as our future work.

Taking a deeper gaze at Table II, it is also indicated that the attack ability of rank-based distinguishers is substantially affected by the group way $g(Z)$. Specifically, taking $g(Z) = Z$ will promote to higher $\widehat{d}_{opt}$ than taking $g(Z) = w_H(Z)$, even under HW leakage function. Interestingly, we find that when categorizing based on HW, the concrete security levels of IPM instances, as reflected by the value of $\widehat{d}'_{opt}$, are consistent with the theoretical analysis proposed in [6]. Specifically, IPM23 demonstrates the highest security level with the third-order bit-probing resilience, while IPM3 and IPM14 exhibit middle-level security levels characterized by $t_b = 2$. Conversely, IPM1 and IPM2 are identified as the least secure instances, with $t_b = t_w = 1$. Moreover, the disparity in bit-probing orders can be illustrated through the magnitudes of optimal efficiency $\widehat{d}'_{opt}$ in statistical analyses.

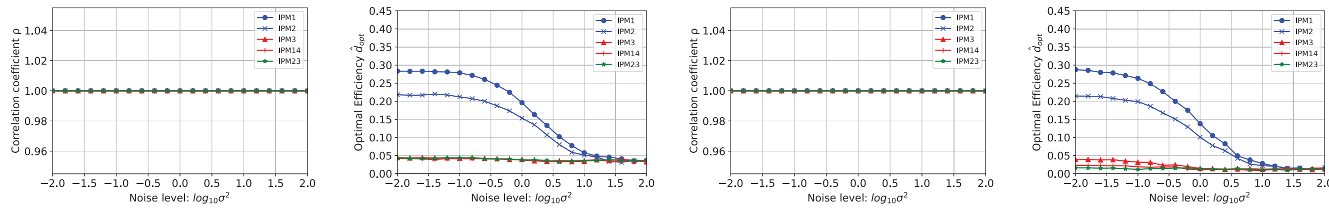
V. EXPERIMENTAL RESULTS

In this section, we validate the statistical analyses through simulated evaluations and real-world attack experiments.

A. Experimental Set-up

The sensitive variable is $Z = Sbox(P \oplus k^*)$ with $Z, P, k^* \in \mathbb{F}_{2^8}$. We assume that Z is protected by two-share IPM. Accordingly, five instances with IPM parameter $a \in \{1, 2, 3, 14, 23\}$ are selected as target implementations, that keep consistent with those examined by a practical evaluation in CHES 2022

[3]Since the optimal efficiency of rank-based distinguishers relies on ranked data $\widehat{C}$, which is related to the specific quantity of measurements, it cannot be derived directly from equations as in Table I.



(a) $\mathsf{Fac}_{\rho_{k*}}$ (b) $\widehat{d}_{opt}$ and $\rho(\mathcal{H}_{opt}(Z), \widehat{C})$ (c) $\mathsf{Fac}_{\rho_{k*}}$ (d) $\widehat{d}_{opt}$ and $\rho(\mathcal{H}_{opt}(Z), \widehat{C})$

Fig. 2. Evolution of three components against the two-share IPM when noise levels increase. The leftmost (resp., rightmost) two sub-figures are under HW (resp., RW) leakage.

paper [9]. Those instances are denoted as IPMx where $x = a$. For example, IPM23 represents the IPM instance with $a = 23$. Since those IPM instances all have two shares, they possess the same first-order security in probing model, namely $t_w = 1$. Following [6], we obtain $d_{\mathcal{D}}^{\perp} \in \{2, 2, 3, 3, 4\}$ for IPMx with $x \in \{1, 2, 3, 14, 23\}$ and thus the bit-probing security order is $t_b = d_{\mathcal{D}}^{\perp} - 1 \in \{1, 1, 2, 2, 3\}$, respectively. Note that IPM1 is exactly BM, while IPM23 is one of the most secure instances of two-share IPM with bit-probing security order t_b up to 3, that can thereby resist third-order attacks in theory [6], [12].

Regarding the leakage function $\mathcal{L}$, We consider four leakage functions in the simulation to align with [16], including two linear types and two non-linear ones, that are listed as below.

- **Hamming Weight (HW):** $\mathcal{L}(z) = w_H(z) = \sum_{i=1}^{n} z[i]$.
- **Randomly Weighted bits (RW):** $\mathcal{L}(z) = w_R(z) = \sum_{i=1}^{n} w[i]z[i]$ with each $w_i \in [-1, 1]$.
- **Strong Non-linear (SN):** $\mathcal{L}(z) = S(Z) = \mathbf{S}(z \bmod 2^4)$, where $\mathbf{S}(x)$ is defined as the S-Box of PRESENT cipher [24].
- **Binary (BN):** $\mathcal{L}(z) = B(z) = \left(\sum_i S(z)[i]\right) \mod 2$.

Note that SN is probably over-estimated in its capability to recombine bits. However it is utilized in [16] to evaluate the attack capabilities of SCD and KWD to demonstrate the advantage of KWD over SCD under non-linear leakages. In this study, we adopt it here to demonstrate the improvement made in newly proposed adjusted SCD and the development of the optimal hypothetical leakage function.

Since two-share IPM is considered in the experiments, we accordingly leverage second-order distinguishers, including 2O-PCD, 2O-SCD, A-2O-SCD and 2O-KWD. The hypothetical leakage function simulated by both 2O-PCD and 2O-SCD satisfies $\mathcal{H}(Z) = w_H(Z)$. Whilst regarding A-2O-SCD, two types of hypothetical leakage functions are utilized, including $\mathcal{H}(Z) = w_H(Z)$ and $\mathcal{H}(Z) = \mathcal{H}_{opt}(Z)$ (as represented in Equation 28). To differentiate, they are denoted as A-2O-SCD$_{HW}$ and A-2O-SCD$_{opt}$, respectively. To showcase the attack efficiency, *Guessing Entropy* (GE) [25] is utilized as a metric.

B. Numerical Evaluations of Optimal Efficiency

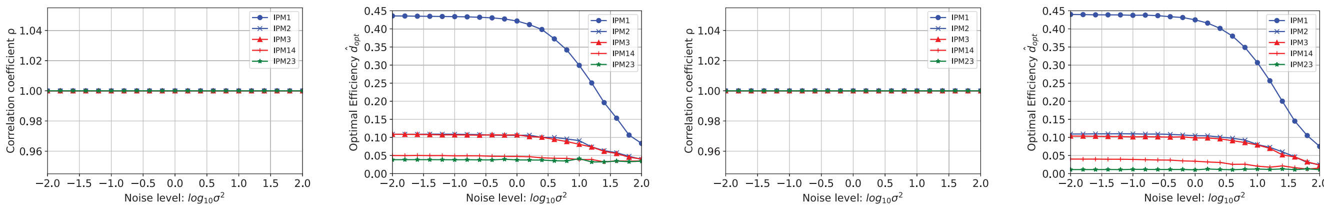
Recall that the distinguishing score $\widehat{d}_{k^*}$ of A-HO-SCD can be decomposed into the product of the correlation coefficient $\mathsf{Fac}_{\rho_{k^*}}$ and the optimal efficiency $\widehat{d}_{opt}$, as presented in Lemma 4. Hence, the values of those two components are critical in determining the attack efficiency of A-HO-SCD. In this subsection, we therefore evaluate their evolution in response to escalating noise levels. It is worth noting that the evolution of $\widehat{d}_{opt}$ indicates the attack efficiency (resp., optimal efficiency) of HO-KWD (resp., HO-SCD) as well. We take $\mathcal{H}(Z) = \mathcal{H}_{opt}(Z)$ derived in Theorem 3 and set the grouping method as $g(Z) = Z$. Regarding simulated leakage functions, we consider $L(Z) = w_H(Z) + N$ and $L(Z) = w_R(Z) + N$ for linear type as well as $L(Z) = S(Z) + N$ and $L(Z) = B(Z) + N$ for non-linear case with $N \sim \mathcal{N}(0, \sigma^2)$. Figure 2 and Figure 3 display the evolution of the correlation coefficient $\mathsf{Fac}_{\rho_{k^*}}$, the optimal efficiency $\widehat{d}_{opt}$ and the attack efficiency $\rho(\mathcal{H}_{opt}(Z), \widehat{C})$ of A-2O-SCD under both linear and non-linear leakage functions, respectively. Since $\widehat{d}_{opt}$ and $\rho(\mathcal{H}_{opt}(Z), \widehat{C})$ are identical in all the simulation settings, we present them by a single figure.

As illustrated in all Figures 2a, 2c, 3a and 3c, the correlation coefficient $\mathsf{Fac}_{\rho_{k^*}}$ remains unaffected by noise levels. Specifically, $\rho(\mathcal{H}_{opt}(Z), \mathbb{E}(\widehat{C}|Z)) = 1$ holds across various two-share IPM instances, under all types of leakage functions, from noise level $\sigma^2 = 0.01$ to $\sigma^2 = 100$. This consequence can be attributed to the fact that $\mathcal{H}_{opt}$ is directly derived from the distribution of $\widehat{C}$, thereby neutralizing the impact of noise when calculating variable probabilities. Hence, $\mathcal{H}_{opt}$ derived in Theorem 3 significantly facilitates A-HO-SCD as it allows the distinguisher to operate at full capacity with its optimal efficiency $\widehat{d}_{opt}$ even in the noisy scenarios under any leakage functions. This enhancement is further evidenced by identical results of $\widehat{d}_{opt}$ and $\rho(\mathcal{H}_{opt}(Z), \widehat{C})$ under both linear and non-linear leakages, as shown in Figures 2b, 2d, 3b and 3d.

Regarding the optimal efficiency $\widehat{d}_{opt}$, it is indicated in Figures 2b, 2d, 3b and 3d that, under considered linear and non-linear leakage functions, the optimal efficiency $\widehat{d}_{opt} > 0$ holds for all the two-share IPM instances across various noise levels. Specifically, as noise level increases, the optimal efficiency for these instances decreases accordingly, gradually converging to approximately 0.035 (resp., 0.015) under HW and SN (resp., RW and BN) leakage. Considering IPMx with $x \in \{3, 14, 23\}$ with bit-probing security order $t_b \geq 2$, the increase in noise level has little effect on the optimal efficiency $\widehat{d}_{opt}$ under HW and RW leakages, indicating that rank-based distinguishers are able to extract key-related information from IP protected implementation, hold promise to compromise SOA and thus achieve successful attacks, even in highly noisy scenarios.

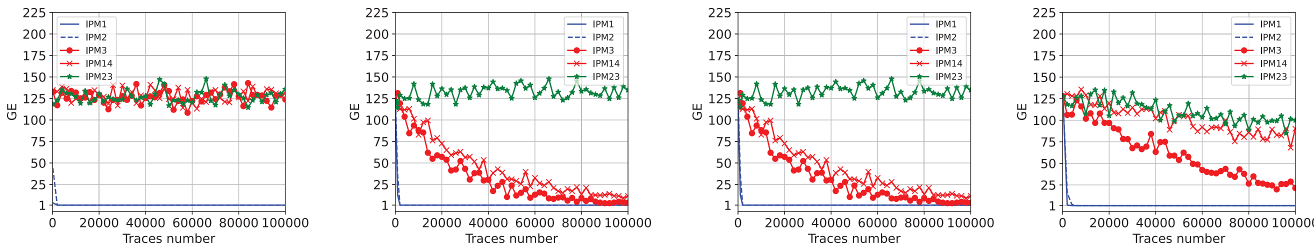
C. Simulated Attack Experiments

Reviewing the statistical analyses in Section IV, two primary conclusions can be attained. On the one hand, the SOA



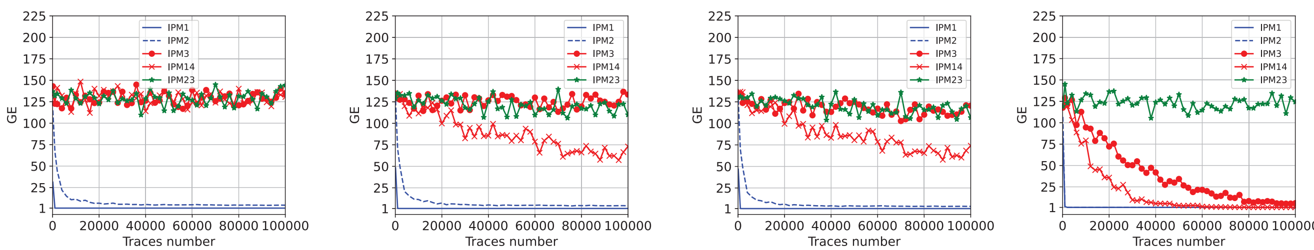
(a) $\mathrm{Fac}_{\rho_{k^*}}$ (b) $\widehat{d}_{opt}$ and $\rho(\mathcal{H}_{opt}(Z), \widehat{C})$ (c) $\mathrm{Fac}_{\rho_{k^*}}$ (d) $\widehat{d}_{opt}$ and $\rho(\mathcal{H}_{opt}(Z), \widehat{C})$

Fig. 3. Evolution of three components against the two-share IPM when noise levels increase. The leftmost (resp., rightmost) two sub-figures are under SN (resp., BN) leakage.



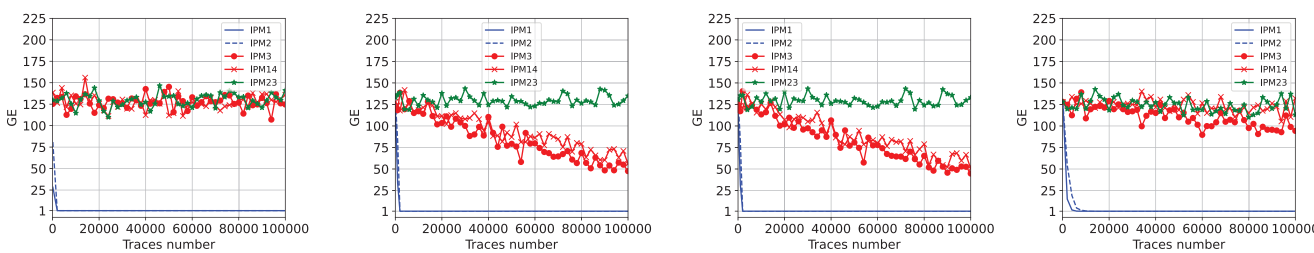
(a) 2O-PCD (b) 2O-SCD (c) A-2O-SCD$_{HW}$ (d) A-2O-SCD$_{opt}$/ 2O-KWD

Fig. 4. GE results of various second-order non-profiling distinguishers under HW leakage when noise level $\sigma^2 = 0.01$.



(a) 2O-PCD (b) 2O-SCD (c) A-2O-SCD$_{HW}$ (d) A-2O-SCD$_{opt}$/2O-KWD

Fig. 5. GE results of various second-order non-profiling distinguishers under RW leakage when noise level $\sigma^2 = 0.01$.



(a) 2O-PCD (b) 2O-SCD (c) A-2O-SCD$_{HW}$ (d) A-2O-SCD$_{opt}$/2O-KWD

Fig. 6. GE results of various second-order non-profiling distinguishers under HW leakage when noise level $\sigma^2 = 1$.

property of IPM is confirmed again according to the statistical analysis on 2O-PCD under HW leakage model. On the other hand, it is observed (somewhat surprisingly) in Lemma 7 that some IPM instances with security order $t_b > t_w$ possibly cannot resist the $(t_w + 1)$th-order rank-based distinguishers, indicating that SOA promisingly loses its effectiveness even under linear leakage from an attack perspective, that is reported for the first time (to the best of our knowledge). The numerical evaluations of the optimal efficiency in Section V-B further verify the latter derivation by showing that $\widehat{d}_{opt}$ convergences to an approximate value (larger than 0) as noise levels increase under various leakage types. In this subsection, we validate those conclusions by simulated attack experiments.

To begin with, we consider linear leakage functions such that $L(Z) = w_H(Z) + N$ and $L(Z) = w_R(Z) + N$ with Gaussian noise $N \sim \mathcal{N}(0, \sigma^2)$. Figure 4 (resp., Figure 5) exhibits the GE results of various distinguishers, including 2O-PCD, 2O-SCD, A-2O-SCD$_{HW}$, A-2O-SCD$_{opt}$ and 2O-KWD from left to right within 100,000 measurements under HW (resp., RW) leakage, when noise level $\sigma^2 = 0.01$. And Figure 6 showcases

TABLE III

GE Results of 2nd-Order Non-Profiling Distinguishers Using 500,000 Traces at Noise Level $\sigma^2 = 0.01$ Under HW Leakage

Distinguishers	IPM1	IPM2	IPM3	IPM14	IPM23
2O-PCD	1.00	1.00	120.67	131.38	122.18
2O-SCD	1.00	1.00	1.00	1.03	157.33
A-2O-SCD$_{HW}$	1.00	1.00	1.00	1.02	157.94
A-2O-SCD$_{opt}$	1.00	1.00	1.00	14.49	39.37
2O-KWD	1.00	1.00	1.00	14.49	39.37

TABLE IV

GE Results of 2nd-Order Non-Profiling Distinguishers Using 500,000 Traces at Noise Level $\sigma^2 = 0.01$ Under RW Leakage

Distinguishers	IPM1	IPM2	IPM3	IPM14	IPM23
2O-PCD	1.00	4.05	127.52	126.12	130.54
2O-SCD	1.00	3.95	104.90	13.17	75.42
A-2O-SCD$_{HW}$	1.00	3.84	59.95	13.48	70.90
A-2O-SCD$_{opt}$	1.00	1.00	1.00	1.00	91.39
2O-KWD	1.00	1.00	1.00	1.00	91.39

TABLE V

GE Results of 2nd-Order Non-Profiling Distinguishers Using 1,000,000 Traces at Noise Level $\sigma^2 = 1.00$ Under HW Leakage

Distinguishers	IPM1	IPM2	IPM3	IPM14	IPM23
2O-PCD	1.00	1.00	127.08	129.37	128.64
2O-SCD	1.00	1.00	1.07	1.60	127.68
A-2O-SCD$_{HW}$	1.00	1.00	1.05	1.34	120.65
A-2O-SCD$_{opt}$	1.00	1.00	2.25	56.14	61.43
2O-KWD	1.00	1.00	2.25	56.14	61.43

the GE results of the same experimental setting under HW leakage yet when noise level increases into $\sigma^2 = 1$. We should underline that the GE results of A-2O-SCD$_{opt}$ and 2O-KWD are identical and thereby displayed together in one figure. In addition, Table III (resp., Table IV) records the GE results using 500,000 traces at noise level $\sigma^2 = 0.01$ under HW (resp., RW) leakage. And Table V provides the GE results using 1,000,000 traces at noise level $\sigma^2 = 1.00$ under HW leakage. Note that *Sbox* function is bijective, rendering the groups $\{Z_k \in \mathbb{F}_{2^n} | \mathbb{E}(\widehat{C}|Z_k)\}$ to be identical for each $k \in \mathbb{F}_{2^n}$. Hence in the simulated attacks, the grouping strategy for both A-2O-SCD$_{opt}$ and 2O-KWD satisfies $g(Z) = (Z \bmod 2^7)$. Consequently, the information of the most significant bit is not utilized by both A-2O-SCD$_{opt}$ and 2O-KWD. For the sake of completeness, we add the simulated attack results of A-2O-SCD with hypothetical leakage function as $\mathcal{H}(Z) = z[8]$ (the most significant bit of the sensitive variable) under HW leakage in Appendix C-B.

Regarding the attack results of 2O-PCD in Figures 4a, 5a and 6a as well as in Tables III, IV and V, it is indicated that SOA plays its role in this case since two-share IPM3, IPM14 and IPM23 with $t_b \geq 2$ are capable of defeating second-order attack under linear leakage (including both HW and RW leakage) in the noisy scenarios. However, considering the attack results (in Table III and V) of both 2O-SCD (in Figures 4b and 6b) and A-2O-SCD$_{HW}$ (in Figures 4c and 6c under HW leakage, it can be observed that both IPM3 and IPM14 with $t_b = 2$ can be successfully attacked at the noise levels $\sigma^2 = 0.01$ and $\sigma^2 = 1$. While considering the attack results (in Table IV) of 2O-SCD (in Figure 5b) and A-2O-SCD$_{HW}$ (in Figure 5c)

TABLE VI

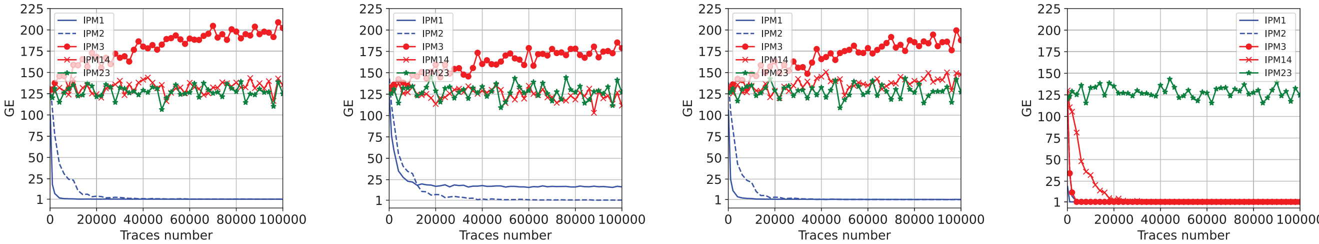
GE Results of 2O-SCD and A-2O-SCD$_{HW}$ Against IPM3 Using Various Amounts of Traces When Noise Level $\sigma^2 = 0.01$

Traces amount	20,000	40,000	60,000	80,000	100,000
2O-SCD	57.10	30.41	13.08	8.80	3.73
A-2O-SCD$_{HW}$	56.14	27.96	10.51	7.56	2.84

under RW leakage, it is indicated that IPM14 can be attacked successfully in these cases, and the GE of IPM3 (resp., IPM23) is decreased to 104.90 and 59.95 (resp., 75.42 and 70.90) using 500,000 traces by 2O-SCD and A-2O-SCD$_{HW}$, respectively, at the noise level $\sigma^2 = 0.01$. Therefore, it is demonstrated that SOA becomes less effective against rank-based distinguishers even under linear leakage. Note that IPM23 is still secure against those two rank-based distinguishers under HW leakage. This is consistent with its d'_{opt} result in Table II that d'_{opt} is too small, being about an order of magnitude smaller than those of IPM3 and IPM14. Recall that the hypothetical leakage function is $\mathcal{H}(Z) = w_H(Z)$ and thus we have $g(Z) = w_H(Z)$ for both 2O-SCD and A-2O-SCD$_{HW}$. Regarding GE results of A-2O-SCD$_{opt}$ and 2O-KWD in Figures 4d, 5d and 6d as well as in Tables III, IV and V, it can be found that all the considered two-share IPM instances are vulnerable at the noise level $\sigma^2 = 0.01$ and $\sigma^2 = 1$ (resp., $\sigma^2 = 0.01$) under HW (resp., RW) leakage, as evidenced by the decreasing trend in the GE across varying degrees. In summary, the simulated attack results further demonstrate that SOA can not guarantee security under linear leakage against rank-based distinguishers, although it contributes to increased attack complexity, necessitating a greater number of measurements.

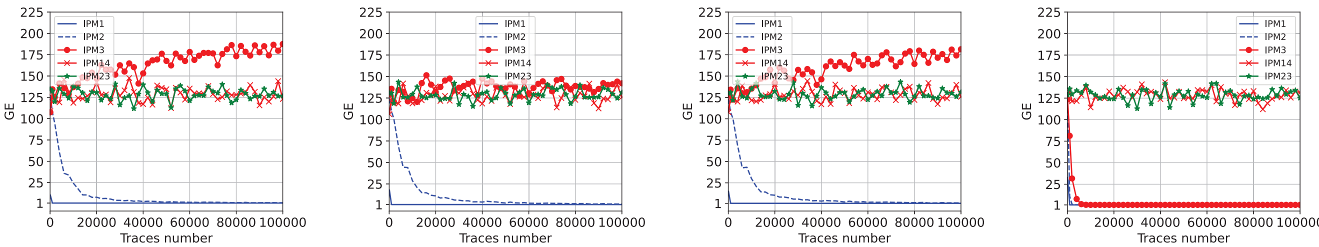
Comparing GE results of 2O-SCD and A-2O-SCD$_{HW}$ in Figures 4, 5 and 6, we can find that they are similar in all the simulation settings. Recall that they utilize the same hypothetical leakage function $\mathcal{H}(Z) = w_H(Z)$. While the difference is whether such hypothetical leakage has undergone a rank transformation. Considering HW leakage, since it is hard to compare their GE results by figures, Table VI records detailed GE values against IPM3 under HW leakage at various numbers of traces. It can be discovered in Tables III, V and VI that GE of A-2O-SCD$_{HW}$ are slightly smaller than those of 2O-SCD across the considered traces amounts. We should underline that this phenomenon holds for all our simulated experiments under HW leakage model, which implies that A-2O-SCD$_{HW}$ is slightly more powerful than 2O-SCD leveraging the same hypothetical leakage function $\mathcal{H}(Z) = w_H(Z)$ under HW leakage model. Regarding RW leakage, Table IV indicates that A-2O-SCD$_{HW}$ and 2O-SCD exhibit comparable capabilities, with the exception of the IPM3 case. Specifically, A-2O-SCD$_{HW}$ demonstrates superior attack ability compared to 2O-SCD in the case of IPM3.

In comparison with A-2O-SCD$_{HW}$, the primary advantage of A-2O-SCD$_{opt}$ lies in its enhanced attack ability against IPM23 under HW leakage. Specifically, the former fails to extract sensitive information from IPM23-protected implementations under HW leakage, as evidenced by the absence of a decrease in GE, even when using 1,000,000 (or 500,000) traces at noise levels of $\sigma^2 = 1$ (resp., $\sigma^2 = 0.01$), as demonstrated



(a) 2O-PCD (b) 2O-SCD (c) A-2O-SCD$_{HW}$ (d) A-2O-SCD$_{opt}$/2O-KWD

Fig. 7. GE results of various second-order non-profiling distinguishers under SN leakage when noise level σ^2 = 0.01.



(a) 2O-PCD (b) 2O-SCD (c) A-2O-SCD$_{HW}$ (d) A-2O-SCD$_{opt}$/2O-KWD

Fig. 8. GE results of various second-order non-profiling distinguishers under BN leakage when noise level σ^2 = 0.01.

in Table V (resp., Table III). In contrast, the latter is able to reduce GE to some extent (implying that some sensitive information can be extracted and leveraged by A-2O-SCD$_{opt}$) under the same experimental conditions, showing that SOA is further compromised due to the development of the optimal hypothetical leakage function. However, in the cases of IPM3 and IPM14, A-2O-SCD$_H$ outperforms A-2O-SCD$_{opt}$ since the hypothetical leakage function $\mathcal{H}(Z) = w_H(Z)$ simulated by A-2O-SCD$_{HW}$ enables $\mathsf{Fac}_{\rho_{k^*}} > \max_{k\in\mathbb{F}_{2^8},k\neq k^*}\{\mathsf{Fac}_{\rho_k}\}$, while the optimal one $\mathcal{H}_{opt}$ leads to $\mathsf{Fac}_{\rho_{k^*}} = \max_{k\in\mathbb{F}_{2^8},k\neq k^*}\{\mathsf{Fac}_{\rho_k}\}$. Additionally regarding RW leakage, A-2O-SCD$_{opt}$ can fully retrieve the true key from implementations protected by IPM2, IPM3 and IPM14 with less than $4,000$, $250,000$, and $60,000$ traces, respectively, as indicated in Figure 5d and Table IV. However, A-2O-SCD$_{HW}$ is unable to break IPM2, IPM3 and IPM14 using $500,000$ traces completely as shown in Table IV, which further confirms the advantage conferred by the improvement of the hypothetical leakage function.

Remark 5: The simulated experimental results of [16] show that 2O-SCD cannot retrieve the key byte from Boolean masked implementations under RW leakage, which contrasts with our findings. One possible reason for this discrepancy is the use of different weight assignments for the individual bits. Recall that [16] randomly assigns weights within the range of $[-1, 1]$ *without providing further clarification on the specific weight values used in their experiments. The specific weights used for our simulated RW leakage are detailed in Appendix C-A.*

Now we consider non-linear cases such that $L(Z) = S(Z) + N$ and $L(Z) = B(Z) + N$ with Gaussian noise $N \sim \mathcal{N}(0, \sigma^2)$. Figure 7 (resp., Figure 8) exhibits the GE results of various distinguishers under SN (resp., BN) leakage at noise level $\sigma^2 = 0.01$, including 2O-PCD, 2O-SCD, A-2O-SCD$_{HW}$, A-2O-SCD$_{opt}$ and HO-KWD. Since *PRESENT Sbox* involves four bits of variables, we set $\mathcal{H}(z) = w_H(z) = \sum_{i=1}^{4} z[i]$ for 2O-PCD, 2O-SCD and A-2O-SCD$_{HW}$, while for A-2O-SCD$_{opt}$ and 2O-KWD, the grouping strategy is $g(Z) = (Z \mod 2^4)$. Since the GE results of A-2O-SCD$_{opt}$ and 2O-KWD keep identical, only a single figure is presented as well.

Regarding the GE results under SN and BN leakage in Figures 7 and 8, respectively, both 2O-SCD and A-2O-SCD$_{HW}$ fail to mitigate IPM instances with $t_b \geq 2$ since the hypothetical leakage function leveraged is not appropriate. And A-2O-SCD$_{HW}$ outperforms 2O-SCD in this case since 2O-SCD even cannot retrieve keys correctly from Boolean masked implementations. Interestingly, A-2O-SCD$_{opt}$ is advanced to break IPM3 and IPM14 (resp., IPM3) due to its improved hypothetical leakage function under SN (resp., BN) leakage. While IPM14 cannot be compromised by A-2O-SCD$_{opt}$ under BN leakage, that may stem from the fact that BN leakage is less informative than SN leakage. Note that IPM23 with $t_b = 3$ is still secure against rank-based distinguishers in these cases. This should be attributed to the fact that only four-bit information is utilized in these scenarios.

Remark 6: We notice that the GE results of 2O-PCD, 2O-SCD and A-2O-SCD$_{HW}$ (resp., 2O-PCD and A-2O-SCD$_{HW}$) against IPM3 exhibit increasing trends under SN (resp., BN) leakage shown in Figure 7 (resp., in Figure 8). Although this phenomenon indicates failed attacks, increasing GE implies that the sensitive information is being leaked and also can be exploited by distinguishers, albeit from an opposing perspective. We should underline that the strong non-linearity character of leakage essentially enables distinguishers (including PCD) to hold promise for effective attacks against IPM instances with higher bit-probing security order, since it increases the numerical degree of leakage that can be leveraged by distinguishers. However, the underlying reason for the

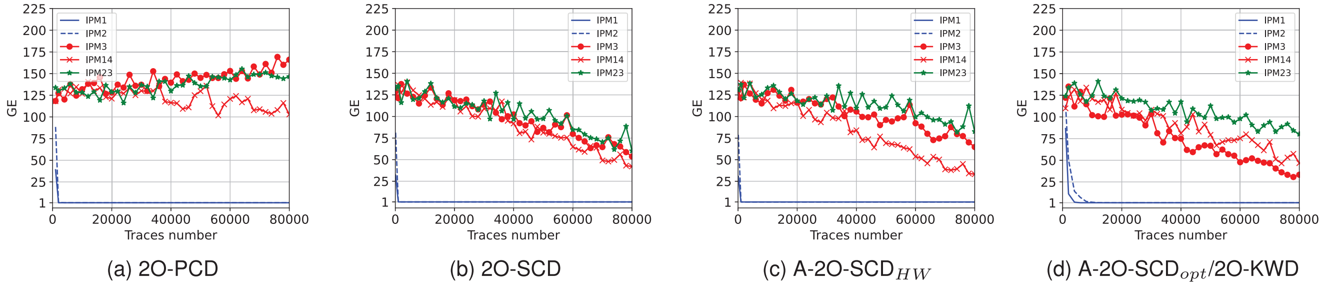


Fig. 9. GE results of various second-order distinguishers in real-world experiments.

increasing trend in GE results remains an open question for future investigation.

D. Real-World Attack Experiments

In this section, we verify the statistical analyses of non-profiling distinguishers against IPM by real-world attack experiments. We evaluate protected AES-128 utilizing the implementation of two-share IPM, that is publicly available as the Artifacts of TCHES 2022 [9]. We use the same implementation strategy and the same experimental platform as in [9] (refer to Acquisition part of Section IV-A).

Figure 9 presents the GE results of 2O-PCD, 2O-SCD, A-2O-SCD$_{HW}$, A-2O-SCD$_{opt}$ and 2O-KWD from left to right. The GE results of A-2O-SCD$_{opt}$ and 2O-KWD keep identical and thus are both displayed in Figure 9d. Note that 2O-PCD exploits the centered product combining function in this paper and the grouping method for A-2O-SCD$_{opt}$ and 2O-KWD is set as $g(Z) = (Z \mod 2^7)$. It is indicated in Figure 9a that SOA confirms effectiveness since IPM instances with $t_b \geq 2$ cannot be mitigated by 2O-PCD, that is consistent with the attack results in [9, Fig. 4(b)] as well as the simulation results in Section V-C. While as shown in Figure 9b, 9c and 9d, all the IPM instances are vulnerable against rank-based distinguishers. Note that physical side-channel leakage in real world is not strictly linear, making it less rigorous to verify the ineffectiveness of SOA—defined under the assumption of linear leakage—against rank-based distinguishers through practical experiments. For instance, the linear regression results of two shares of IPM23 in our setting, as shown in Figure 11 of Appendix D, imply that physical leakage primarily comprises a dominant linear component along with a minor non-linear part, as similarly observed in [5, Fig. 2]. The significant dominance of linear component favors the SOA of IPM, thereby enabling IPM instances with $t_b \geq 2$ to resist 2O-PCD. However, the minor non-linear component in the leakage enhances the capability of rank-based distinguishers to compromise IPM instances. For example, while both 2O-SCD and A-2O-SCD$_{HW}$ cannot attack IPM23 in the simulated HW experiments as presented in Figure 4, they succeed in decreasing the GE of IPM23 to 80 and below. IPM23 also exhibits greater vulnerability to A-2O-SCD$_{opt}$ and 2O-KWD than observed in the simulated HW experiments. Nevertheless, the comparative attack results in Figure 9 underscore the superiority of rank-based distinguishers in extracting information from physical leakage.

Comparing GE results of rank-based distinguishers in Figure 9b, 9c and 9d, they present similar attack ability, having each advantage against various IPM instances. This is not consistent with the attack results of simulated experiments under both linear and non-linear leakages. One reason is that the actual leakage function of physical device is different, as analyzed above. And the underlying reason is that A-HO-SCD and the optimal hypothetical leakage function derived in Theorem 3 are developed to promote $\mathsf{Fac}_{\rho_{k^*}} = 1$, rather than maximizing $(\mathsf{Fac}_{\rho_{k^*}} - \max_{k \in \mathbb{F}_{2^8}, k \neq k^*} \{\mathsf{Fac}_{\rho_k}\})$. The latter is the key for successful side-channel attacks. However, it is worth noting that A-HO-SCD has been shown to outperform HO-SCD under both linear and non-linear leakages by simulated experiments. In particular, the optimal hypothetical leakage function enables A-HO-SCD to compromise IPM23 under HW leakage and possess a significant advantage under RW, SN and BN leakages.

VI. Discussion and Conclusion

A. SOA and the Security of IPM

As introduced in Section I-A, SOA can be interpreted from three perspective. Specifically, SOA still holds from the viewpoints of mutual information and theoretical probing models. However, as demonstrated in this study, SOA fails to provide the expected protection for IP-masked implementations against rank-based distinguishers. Therefore, a more fine-grained investigation into the protection capability of IPM (including SOA), as well as other high-algebraic masking schemes, is required. This includes, for example, the analysis of adversaries' leakage extraction capabilities to better understand the security of these schemes.

Although this paper highlights that SOA can be compromised by rank-based distinguishers, it is important to emphasize that IPM with bit-probing security order $t_b \geq 2$, remains significantly more secure than BM, requiring substantially more measurements for a successful attack (e.g., up to 1,000 times). This is supported by both simulated and practical experiments presented in this study. Furthermore, IPM23 with $t_b = 3$ demonstrates exact security in certain scenarios. Moreover, the concrete security levels of IPM instances considered in this paper follow the theoretical quantification framework proposed in [6].

B. Adjusted SCD and Its Optimal Hypothetical Leakage Function

Adjusted SCD is devised to enable $\rho(\widehat{\mathcal{H}(Z)}, E(\widehat{C}|Z)) = 1$ to maximize d_{k^*} and thus remove the rank transformation of $\mathcal{H}(Z)$. This suggests two theoretical implications. On the one hand, if the hypothetical leakage function is defined as $\mathcal{H}(Z) = \mathcal{H}_{opt}(Z) = E(\widehat{C}|Z)$ and accordingly d_{k^*} is optimized as $\widehat{d}_{opt}$, then Adjusted SCD should achieve better results than SCD in terms of the value of d_{k^*}. Hence, as demonstrated in the simulated experiments, Adjusted SCD using the optimal hypothetical leakage function (as formalized in Equation 22) shows a significant improvement over SCD in the most simulated scenarios.

However, on the other hand, the maximum d_{k^*} (such that $d_{k^*} = \widehat{d}_{opt}$) does not suffice to indicate the optimal attack efficiency, as declared in Remark 1. For instance, since the optimal hypothetical leakage function derived in Theorem 3 makes $\mathsf{Fac}_{\rho_k} = 1$ for every $k \in \mathbb{F}_{2^8}$ (including k^*), 2O-SCD outperforms A-2O-SCD$_{opt}$ for a few cases (e.g., IPM3 and IPM14 under HW leakage) where $\mathcal{H}(Z)$ enables $\mathsf{Fac}_{\rho_{k^*}} > \max_{k \in \mathbb{F}_{2^8}, k \neq k^*} \{\mathsf{Fac}_{\rho_k}\}$. In addition, the attack effectiveness of A-2O-SCD$_{opt}$ is equivalent to 2O-KWD in theory.

Therefore, if the hypothetical leakage function enlarges the gap between $\mathsf{Fac}_{\rho_{k^*}}$ (resp., $\mathsf{Fac}_{\widehat{\rho}_{k^*}}$) and $\max_{k \in \mathbb{F}_{2^n}, k \neq k^*} \{\mathsf{Fac}_{\rho_k}\}$ (resp., $\max_{k \in \mathbb{F}_{2^n}, k \neq k^*} \{\mathsf{Fac}_{\widehat{\rho}_k}\}$) as much as possible, Adjusted SCD (resp., SCD) can be further enhanced to achieve better attack effectiveness than KWD. We should underline that it is indeed challenging to theoretically compare the attack capabilities of Adjusted SCD and SCD without assuming $\mathcal{H}(Z) = \mathcal{H}_{opt}(Z) = E(\widehat{C}|Z)$ for the hypothetical leakage function. However although without theoretical guidelines, experiments considering Adjusted SCD and SCD with other hypothetical leakage functions are included for the sake of completeness. The results demonstrate that Adjusted SCD exhibits comparable, and in most cases slightly superior, attack capability to the SCD when employing the same hypothetical leakage function. Moreover, in certain instances,such as IPM3 under RW leakage and IPM1 under SN leakage, Adjusted SCD substantially outperforms SCD in terms of attack effectiveness.

In conclusion, Adjusted SCD (resp., SCD) can be further enhanced by developing a hypothetical leakage function that maximizes the gap between $\mathsf{Fac}_{\rho_{k^*}}$ (resp., $\mathsf{Fac}_{\widehat{\rho}_{k^*}}$) and $\max_{k \in \mathbb{F}_{2^n}, k \neq k^*} \{\mathsf{Fac}_{\rho_k}\}$ (resp., $\max_{k \in \mathbb{F}_{2^n}, k \neq k^*} \{\mathsf{Fac}_{\widehat{\rho}_k}\}$), thereby yielding better attack results compared to KWD. Notably, we do not theoretically assert that Adjusted SCD is a more advanced distinguisher than SCD in terms of attack effectiveness. Nevertheless, experimental results indicate that Adjusted SCD shows considerable promise as a potential replacement for SCD in most scenarios, particularly given that Adjusted SCD can achieve $\mathcal{H}(Z) = \mathcal{H}_{opt}(Z) = E(\widehat{C}|Z)$ to maximum d_{k^*}.

C. Other Non-Profiling Distinguishers

In contrast to PCD, rank-based distinguishers (including SCD, Adjusted SCD and KWD) can compromise SOA of IPM since the ranked data offer more informative insights than the original image, enabling that $\sigma(\mathbb{E}(\widehat{C}|Z)) > 0$, as analyzed in Section IV-C. In addition to rank-based distinguishers, there exist other non-profiling attack techniques capable of exploiting non-linear links and retrieving keys from masked implementations, such as JMR method [15], higher-order moment-based attack [12] and non-profiling Deep-Learning (DL) based approach [26]. Specifically, JMR method makes hypothesis for each share to approach the joint moment distribution of empirical measurements by non-linear regression, while the higher-order PCD in [12] is constructed to exploit higher-order moments (much higher than 2) in extracting key-dependent leakages. Notably, both of them share the same spirit in exploiting higher-order dependence in leakage rather than only at the order 2. Another different approach is the DL-based method [26], which utilizes DL training to evaluate the consistency of partitions divided by various hypothetical intermediate variables. Since they are not based on ranked data and the attack effectiveness does not directly depend on $\sigma(\mathbb{E}(C|Z))$ or $\sigma(\mathbb{E}(\widehat{C}|Z))$, we exclude the investigation of their attack capabilities against IPM from the scope of this work. However, exploring and comparing these methods against IPM will be part of our future research.

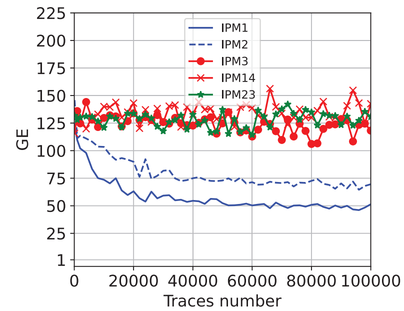
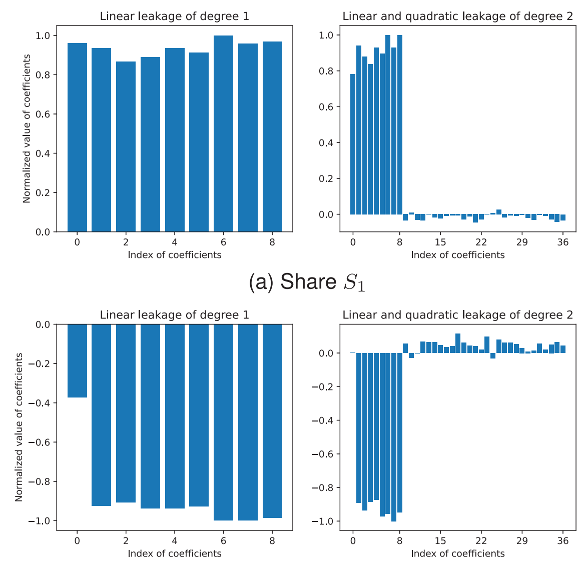


Fig. 10. GE results of A-2O-SCD using $\mathcal{H}(Z) = z[8]$ under HW leakage at the noise level $\sigma^2 = 0.01$.



(a) Share S_1

(b) Share S_2

Fig. 11. Linear regression results of shares S_1 and S_2 for IPM23 in our experimental setting.

D. Conclusions and Perspectives

In this paper, we revisit SOA of IPM from a non-profiling attack perspective. First, we delineate a theoretical connection between HO-SCD and HO-KWD through statistical analysis. Specifically, we demonstrate that HO-SCD is a more generic distinguisher that encompasses HO-KWD. We also propose a new side-channel distinguisher called A-HO-SCD, which shall enhance HO-SCD to maximize attack efficiency. Second, SOA is re-examined by lens of non-profiling distinguishers. We construct the optimal form of A-HO-SCD to achieve the optimal efficiency against IPM, meanwhile bridging the efficiency gap with HO-KWD. Furthermore, we demonstrate that, unlike HO-PCD, rank-based distinguishers show potential in breaking SOA of IPM from an attack perspective even under linear leakage (verified by simulated and practical experiments). This finding, which contradicts conventional understanding, is reported for the first time to the best of our knowledge.

As analyzed above, A-HO-SCD and its hypothetical leakage function are developed to maximize $\mathsf{Fac}_{\rho_{k^*}}$ in this study, rather than $(\mathsf{Fac}_{\rho_{k^*}} - \max\limits_{k\in\mathbb{F}_{2^8}, k\neq k^*}\{\mathsf{Fac}_{\rho_k}\})$. The former is an significant indicator, while the latter is the crucial for successful attacks, which will be further explored in our future work. Regarding SOA against non-profiling distinguishers, it remains an open problem how rank-based distinguishers extract higher-order moments under linear leakages, which will be our focus of research as well. Despite this, current investigation is sufficient to demonstrate that the side-channel resistance guaranteed by SOA is vulnerable even in the ideal linear leakage model. At last, our investigations into non-parametric distinguishers enhance the tool-set for the non-profiling class of SCA, aiding in the evaluation of the practical side-channel security.

Appendix A
Proof of Theorem 1

Proof: According to Equation 25, we have

$$\begin{aligned}\mathbb{E}(C|Z=z) &= \mathbb{E}(w_H(z\oplus a\cdot M)w_H(M)|Z=z)\\ &\quad -\frac{n}{2}\mathbb{E}(w_H(z\oplus a\cdot M)|Z=z).\end{aligned}\tag{35}$$

Applying Property 1, we have

$$\begin{aligned}&\mathbb{E}(w_H(z\oplus a\cdot M)w_H(M)|Z=z) = \mathbb{E}((w_H(z)+w_H(a\cdot M)\\ &\quad -2w_H(z\wedge(a\cdot M)))w_H(M)|Z=z) = \frac{n}{2}w_H(z)\\ &\quad +\mathbb{E}(w_H(a\cdot M)w_H(M))\\ &\quad -2\mathbb{E}(w_H(z\wedge(a\cdot M))w_H(M)|Z=z).\end{aligned}\tag{36}$$

And $\mathbb{E}(w_H(z\wedge(a\cdot M))w_H(M)|Z=z)$ satisfies

$$\begin{aligned}&\mathbb{E}(w_H(z\wedge(a\cdot M))w_H(M)|Z=z)\\ &=\mathbb{E}\left(\sum_{i=1}^{n}(z[i]\wedge(a\cdot M)[i])w_H(M)|Z=z\right)\\ &=\sum_{i=1}^{n}z[i]\mathbb{E}((a\cdot M)[i]w_H(M)) = \sum_{i=1}^{n}z[i]\sum_{j=1}^{n}P_{i,j}.\end{aligned}\tag{37}$$

Considering $\mathbb{E}(w_H(z\oplus a\cdot M)|Z=z)$, also applying Property 1, we have

$$\begin{aligned}&\mathbb{E}(w_H(z\oplus a\cdot M)|Z=z) = \mathbb{E}(w_H(z))+w_H(a\cdot M)\\ &\quad -2w_H(z\wedge(a\cdot M)|Z=z) = w_H(z)\\ &\quad +\frac{n}{2}-2\mathbb{E}(w_H(z\wedge(a\cdot M))|Z=z) = w_H(z)+\frac{n}{2}\\ &\quad -2\mathbb{E}\left(\sum_{i=1}^{n}z[i]\wedge(a\cdot M)[i]|Z=z\right)=\frac{n}{2}.\end{aligned}\tag{38}$$

Hence, combining Equations 35, 36, 37, and 38 leads to Equation 26, which completes the proof. □

Appendix B
Proof of Theorem 3

Proof: Note that $\mathbb{E}(\widehat{C}) = \frac{m+1}{2}$. By Lemma 2, we have

$$\begin{aligned}\sigma^2(\mathbb{E}(\widehat{C}|g(Z)=y)) &= \mathbb{E}(\mathbb{E}^2(\widehat{C}|g(Z)=y))\\ &\quad -\mathbb{E}^2(\mathbb{E}(\widehat{C}|g(Z)=y))\\ &= \frac{m^2}{4}(\mathbb{E}((\vec{p^*}\cdot\vec{p^y})^2)-1)\\ &\quad +\frac{m}{2}(\mathbb{E}(\vec{p^*}\cdot\vec{p^y})-1).\end{aligned}\tag{39}$$

Since $p_i=\sum_{y\in\mathcal{Y}}P[g(Z)=y]p_i^y$, we have

$$\begin{aligned}\mathbb{E}(\vec{p^*}\cdot\vec{p^y}) &= \sum_{y\in\mathcal{Y}}P[g(Z)=y]\sum_{i=1}^{v}p_i^*p_i^y\\ &=\sum_{i=1}^{v}p_i^*\sum_{y\in\mathcal{Y}}P[g(Z)=y]p_i^y\\ &=\sum_{i=1}^{v}p_i^*p_i=1.\end{aligned}\tag{40}$$

In addition, $\sigma^2(\widehat{C})$ satisfies

$$\begin{aligned}\sigma^2(\widehat{C}) &= \sum_{i=1}^{v}p_i(\widehat{c}_i-\mathbb{E}(\widehat{C}))^2 = \sum_{i=1}^{v}p_i\left(\frac{m}{2}p_i^*+\frac{1}{2}-\frac{m+1}{2}\right)^2\\ &=\frac{m^2}{4}\sum_{i=1}^{v}p_i((p_i^*)^2-2p_i^*+1)\\ &=\frac{m^2}{4}\sum_{i=1}^{v}p_i(p_i^*)^2-\frac{m^2}{4}.\end{aligned}\tag{41}$$

Hence, combining Equations 39, 40, and 41 leads to Equation 31, which completes the proof. □

Appendix C
Other Simulated Experiments

A. Simulated Weights in RW Case

The simulated weights in our RW case in Section V-C is $\vec{w}=[0.96,-0.77,-0.62,0.49,-0.34,0.81,0.46,-0.8]$ from the least to the most significant bit.

B. Ge Results Under HW Leakage

The GE Results of A-2O-SCD using $\mathcal{H}(Z)=z[8]$ under HW leakage is shown in Figure 10 by using 100,000 measurements at the noise level $\sigma^2=0.01$. In particular, GE will be further decreased to 46.09, 64.96, 69.31, 174.42 and 122.03 for IPM1, IPM2, IPM3, IPM14 and IPM23, respectively, by exploiting 1,000,000 traces. Therefore, as demonstrated in conjunction with Figure 10, it can be inferred that a single bit provides insufficient information for key retrieval.

APPENDIX D
LINEAR REGRESSION OF IPM23 SHARES

See Figure 11

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